

Beliefs, Reputation, and Barriers to Entry in Online Labor Markets^{*}

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Abstract

Hundreds of millions of workers in developing countries seek digital jobs online but face entry barriers without an established reputation. Novice workers could offset this by lowering initial wages, yet few do. Our baseline survey points to two explanations: workers believe employers interpret low wages as low quality signals and are uncertain about their own abilities. We conduct two field experiments on a leading global freelancing platform to examine how these beliefs shape worker outcomes. In the demand-side experiment, we randomize wage offers by novice workers to 703 jobs and find that employers respond favorably to lower wage offers from novices. In the supply-side experiment with 481 novice workers, we randomly provide them with accurate information about employer responses and their performance. Correcting workers' beliefs increases their willingness to lower wages. Consistent with reputation models, effects are driven by high-ability novices with high returns to reputation once these frictions are removed. Simulations show that without external intervention, worker learning about employer responses is slow and costly. Our findings highlight that information interventions can help workers in developing countries overcome reputation barriers and accelerate talent discovery in online labor markets.

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1 Introduction

Inexperienced workers often struggle more than experienced ones to find jobs, particularly in low- and middle-income countries (LMICs).¹ A key challenge is their inability to credibly signal their skills to employers, particularly when their qualifications are difficult to evaluate or verify (Abebe et al., 2021; Bertrand et al., 2024). This problem is especially pronounced in the rapidly expanding online labor market, where hundreds of millions of workers from these countries seek digital job opportunities on global freelancing platforms (Datta et al., 2023). Although these platforms expand access to opportunities, information frictions about worker ability create substantial entry barriers for inexperienced workers without prior online job histories (Agrawal, Lacetera and Lyons, 2016; Hilbert and Lu, 2020). Because employers cannot observe worker ability prior to hiring, they favor workers with established reputations, creating a “reputation trap” for inexperienced workers: they need jobs to build reputation but need reputation to secure jobs. Classic reputation models (Tirole, 1988; Stiglitz, 1989) suggest that inexperienced workers can overcome this problem through “introductory pricing”: offering lower wages initially to attract employers and raising them once reputation is established. Yet in practice, inexperienced workers rarely lower wages to build reputation. Why do they refrain from such reputation investment?

We study this question using detailed platform and survey data from LMIC freelancers, together with two randomized controlled trials on a leading global freelancing platform. We first document suggestive evidence that this pattern may reflect workers’ beliefs about employer responses or their own abilities. Workers may hesitate to lower wages because they believe employers interpret low prices as signals of low quality.² Alternatively, workers who are uncertain about their own ability cannot predict their future performance or the reputation they will establish, making them reluctant to forgo initial earnings for future returns to reputation.

The first experiment tests whether employers truly interpret low wage offers from inexperienced freelancers as signals of low quality. We randomize wage offers by inexperienced (novice) and experienced (veteran) freelancers and compare employer responses. The second experiment examines how novices’ beliefs shape wage offers. We randomly provide novices with information about employer responses and/or their own ability, and measure the effects on subsequent wage offers. Our results show that novices’ inaccurate beliefs about employer responses and their own ability deter reputation investment and market entry, especially among high-ability workers. These frictions not only distort talent allocation in the online

¹For example, in some regions, they are three times more likely to be unemployed than the general population (Alfonsi et al., 2020).

²This pattern has been well documented in product markets where consumers infer lower quality from lower prices when evaluating unfamiliar goods. See Rao and Monroe, 1989; Wathieu and Bertini, 2007; Erdem, Keane and Sun, 2008; Ashraf, Jack and Kamenica, 2013.

market, but are especially costly in LMICs, where limited local employment opportunities make access to online work particularly valuable for skilled workers.

We focus on data entry freelancers from LMICs on one of the largest global freelancing platforms. On the platform, employers post jobs with stated budgets, and freelancers apply by submitting wage bids. We define *undercutting* as freelancers bidding below the posted budget. When evaluating applicants, employers observe each freelancer’s wage bid and reputation, represented by the number of completed jobs and job success rate on the platform. We combine comprehensive platform data on freelancers’ profile characteristics, job histories, and applications with original survey and performance data collected from 481 novice freelancers across 37 LMICs hired for a standardized data entry task on the platform.

Our descriptive analysis reveals a puzzle. Reputation generates large returns to future earnings, and these returns are heterogeneous: public employer reviews increase hiring chances for high-performing novices but shut out low-performing ones (Pallais, 2014; Fazio, Freund and Novella, 2025). High-ability novices therefore stand to gain the most from securing initial jobs and building reputation early. This is especially important from an efficiency standpoint: novices in our data perform no worse than experienced workers with established reputations, suggesting the reputation trap may generate talent misallocation. Yet despite intense competition, wage bids vary little, even among novices with limited outside employment options.³ Our original data points to two explanations. First, although novice performance varies substantially, workers have limited knowledge of their abilities relative to others. Without knowing her type, a novice cannot predict whether she will perform well enough to recoup the initial earnings sacrifice, because a poor reputation may prevent her from securing future jobs. Second, novices believe that employers interpret low wage bids as signals of low quality, further deterring undercutting.

Motivated by these descriptive findings, our first experiment examines whether employers interpret low wage bids from novices as negative quality signals and thus whether undercutting is a viable entry strategy. To generate exogenous variation in wage bids, we create experimental profiles for novice and veteran freelancers with comparable characteristics, randomize their bids across 703 real jobs on the platform, and track employer responses.

The results are contrary to workers’ baseline beliefs. Although novices are less likely than veterans to receive employer responses, low wage bids significantly increase employers’ likelihood of viewing the application and contacting freelancers in both groups. Crucially, employers are more responsive to low bids from novices than from veterans: novice callback rates increase by 54 percent from a baseline of 3.41 percentage points, compared to a 16 percent increase

³Our sample represents a substantial underemployed labor force with limited job opportunities. Nearly half report no offline work, and more than 80% have no completed jobs on this or any other freelancing platform. For these workers, overcoming initial entry barriers and building a reputation to secure digital jobs is especially valuable.

from a baseline of 5.26 percentage points for veterans. These patterns are inconsistent with employers interpreting low wages as negative quality signals. Instead, employers appear to view lower bids by novices as a willingness to invest in reputation. One concern is that even if undercutting helps novices secure jobs, the future returns to reputation may not offset the initial earnings loss. Our back-of-the-envelope calculations suggest otherwise: novices who consistently undercut can more than double their net earnings within the first year. Taken together, these findings suggest that undercutting can help novices overcome the reputation trap, raising the question of why so few workers adopt this strategy.

Our second experiment therefore examines how novices' beliefs about employer demand and their own ability affect wage bidding behavior. The experiment is informed by [Tirole \(1988\)](#), in which new entrants offer introductory pricing as an investment in reputation to overcome initial information frictions about their quality. In the original model, a separating equilibrium can emerge where high-ability novices offer low wages initially to build reputation for future transactions, while low-ability novices do not. Low wage offers by high-ability novices increase their chances of being hired and improve match quality. We show that the separating equilibrium breaks down when novices hold inaccurate beliefs about employer responses and their own ability. This framework yields testable predictions: correcting beliefs about demand encourages undercutting broadly, while correcting beliefs about own ability induces undercutting by high-ability novices.

We test these predictions in an experiment with 481 novice freelancers hired for the baseline data entry task, for whom we have measured their performance and beliefs about own ability and demand responses. Identifying the mechanisms using observational data alone is difficult because ability, beliefs, and wage bids are jointly determined. Our experiment generates exogenous shifts in beliefs by randomly providing novices with private information about their task performance relative to other novices and/or information about employer responses to wage bids. We then track their wage bids for a real job referred to them by us on the platform.⁴ Leveraging baseline and endline surveys, we confirm that the treatments substantially shift novices' beliefs through the intended channels, without affecting perceptions about other aspects of the job application process.

We find that correcting novices' beliefs significantly increases their willingness to offer low wage bids, but through different channels. When novices learn that employers do not treat low bids as negative quality signals, the share who undercut nearly quadruples relative to the control group, from 5.5 percent to 20.5 percent, with no difference between high- and low-ability workers. By contrast, information about own performance increases the share of workers who undercut to 14.4 percent. This effect is concentrated among high-ability

⁴We use this referral approach because job applications are not publicly observable on the platform. It allows us to observe application behavior while preserving workers incentives to appeal to prospective employers.

novices, especially those with more accurate baseline beliefs about employer demand. This pattern suggests that high-ability workers understand which strategies are effective but are uncertain about their own standing; receiving credible information about their ability induces them to invest in reputation through lower initial wages. When workers receive both types of information, we observe separation in wage bids: high-ability novices become five times more likely to undercut relative to the control group, while low-ability novices become only twice as likely. Moreover, we find no evidence that workers who undercut are more likely to renegotiate wages or produce lower-quality output. Taken together, the results show that misperceptions about employer demand and uncertainty about own ability both deter novices from investing in reputation. Information about employer responses encourages more novices to lower wage bids to secure initial jobs, while information about own ability primarily induces high-ability novices to invest in reputation through undercutting, improving talent allocation as in [Tirole \(1988\)](#).

The demand- and supply-side experiments show that inaccurate beliefs prevent novice workers from offering lower initial bids to overcome entry barriers. For policymakers, an important question is whether these belief frictions naturally dissipate with experience on the platform or whether external interventions are needed. Using estimates from both experiments, we simulate a Bayesian learning process to assess the time and monetary costs for perfectly rational workers to learn employer responses by experimenting with different bidding strategies. Even under systematic experimentation and perfect updating, learning is slow and costly, which help explain why these frictions may persist: a median worker with moderately inaccurate beliefs would require roughly 200 job applications and incur about \$270 in application fees.

We also draw on recent evidence to argue that similar supply-side frictions exist among online freelancers in other service categories and in traditional offline labor markets in developing countries. More broadly, these findings help explain why skill training programs for inexperienced jobseekers alone often have limited effects on employment outcomes. While such programs improve workers' skills, novices still face barriers in credibly signaling those skills to employers and acquiring the initial reputation needed to secure future jobs. Our results instead point to the importance of external information provision (e.g., through mentorship programs) to facilitate market access and the growth of high-ability workers.

This paper bridges three fundamental issues that constrain young jobseekers in LMICs: workers' limited ability to signal their skills, their lack of information about the labor market and their own abilities, and the central role of initial reputation in securing jobs - particularly in online labor markets. We study these frictions in the increasingly important context of global freelancing platforms. Employers on these platforms rely heavily on workers' job histories to infer quality ([Pallais, 2014](#); [Agrawal, Lacetera and Lyons, 2016](#); [Barach and Horton,](#)

2019), and alternative signals available to novice workers, such as micro-credentials or advertising, have limited effectiveness (Kässi and Lehdonvirta, 2022; Hannane, 2024). Building on classic reputation models (Tirole, 1988; Stiglitz, 1989), we provide causal evidence that novice workers can compensate for missing reputation by initially lowering wages. However, they under-adopt this strategy because of the belief frictions documented above. While prior work emphasizes upskilling freelancers in LMICs and finds limited effects on platform success (Baptista, Freund and Novella, 2023; Das et al., 2024; Fazio, Freund and Novella, 2025), our results show that when workers cannot credibly signal their skills, correcting belief frictions is essential for adopting strategies that overcome entry barriers.

Second, we contribute to growing body of work documenting jobseekers’ inaccurate beliefs as a barrier to employment (see Caria et al., 2024 for a review). These beliefs are central because they shape job search strategies and, in our setting, determine whether workers are willing to sacrifice current earnings to build reputation. Prior work emphasizes two types of misbeliefs. The first concerns overly optimistic expectations about wages (Banerjee and Sequeira, 2023; Alfonsi, Namubiru and Spaziani, 2024; Kelley, Ksoll and Magruder, 2024) or hiring probabilities (Chakravorty et al., 2024; Abebe et al., 2025; Bandiera et al., 2025), which can lead jobseekers to reject viable opportunities. We identify a new mechanism: misbeliefs about how employers interpret low wages as signals of worker quality. We experimentally show that even though novices are willing to work for less to prove themselves, pessimistic beliefs about employer responses deter them from adopting effective wage bidding strategies. The second concerns workers’ imperfect information about their own abilities (Bassi and Nansamba, 2022; Carranza et al., 2022; Kiss et al., 2023). Correcting these misperceptions helps jobseekers target roles aligned with their strengths. We extend this insight by embedding it in a reputation model and show that performance feedback induces high-ability novices to lower wages to build reputation, generating separation in bidding behavior and potentially accelerating talent discovery on the platform.

Finally, we contribute causal evidence to the theoretical literature on prices as signals of quality, from both sides of a labor market. Theory predicts that market conditions determine whether sellers signal high quality through *high* prices (Wolinsky, 1983; Milgrom and Roberts, 1986; Judd and Riordan, 1994) or *low* prices (Tirole, 1988; Stiglitz, 1989). Empirical work has focused almost entirely on product markets. Empirical tests of these predictions in labor markets are scarce because worker types and beliefs are inherently unobservable. Roussille (2024) shows suggestive evidence that employers interpret higher wage requests as signals of higher ability on a US online labor platform. Closest to our work, Huang, Li and Zuo (2025) examine how novice domestic-service workers in China use prices to signal quality, relying on initial worker characteristics as proxies for unobserved types. We advance this literature by collecting novel, directly measured data on workers’ true abilities and their beliefs, and by experimentally varying both wage bids and information. This approach al-

allows us to show that novice workers and employers hold systematically misaligned beliefs about how wages signal quality, and that these belief gaps are costly to correct in a market characterized by high search costs and intense competition.

The remaining sections are organized as follows. Section 2 provides background on the online freelancing platform and online labor markets more broadly. Section 3 describes the data collection and presents the motivating facts. Section 4 outlines the demand-side experimental design and present the results. Section 5 introduces the supply-side experimental design, motivated by a reputation model in the vein of [Tirole \(1988\)](#). Section 6 presents the results. Section 7 discusses policy implications for our findings and their generalizability to other labor market settings. Section 8 concludes.

2 Online Freelancing Markets

Online freelancing markets have become an important component of the global workforce. Recent estimates suggest that between 154 and 435 million people engage in online gig work, with the majority based in LMICs ([Datta et al., 2023](#)). These platforms connect workers with employers to perform and deliver remote, one-time tasks ranging from simple data entry to complex software development and graphic design. Workers are paid either per-project (fixed payment) or on an hourly basis.

For this study, we focus on fixed-payment, data entry jobs from one of the world’s largest online freelancing platforms. This setting is well-suited to studying how novice workers from LMICs enter online labor markets for several reasons. First, data entry is highly prevalent: according to the World Bank, it represents the second most popular job category, accounting for 23% of total jobs after software development and technology ([Datta et al., 2023](#)). Second, these jobs are accessible to new entrants from LMICs, requiring less advanced skills and serving as a common entry point to the platform. Third, fixed-payment contracts dominate initial employment: 74% of data entry workers in our sample receive fixed-payment contracts in their first job. Together, these characteristics make this an ideal context for examining the early platform experiences of LMIC workers.

To understand how workers navigate this market, we outline the job matching process below. While we focus on a specific platform, this process shares key features with other major platforms. First, employers post a public job listing specifying the task description, required skill level, expected duration, and budget (either fixed payment or an hourly rate). Second, workers either apply directly to the posting or receive an invitation from the employer; for new workers, direct applications are the most common pathway. Third, to apply, workers submit a wage bid, a cover letter, and pay an application fee.⁵ Wage bids can fall above or below the

⁵According to the platform, the application fee serves as a screening mechanism to reduce congestion by

posted budget, provided they meet minimum requirements (\$5 for fixed-payment projects, \$3 per hour for hourly projects). Bids are visible to employers, not to other applicants.

Once applications are submitted, employers review applicants through a dashboard displaying basic information: profile title, number of completed jobs and earnings, job success rates, self-reported skills, and wage bid. As shown in Appendix Figure A1, the number of completed jobs, job success rates, and wage bid are the most salient features. Employers can also click on individual applications to review cover letters and full freelancer profiles, which typically include self-reported credentials and detailed job history. To move forward, employers contact workers through the platform’s messaging system to either initiate contracts or request additional information. Although wage negotiation is possible at this stage, it rarely occurs (Barach and Horton, 2019), and workers are typically paid their proposed wage bid. Once a contract is signed, the job automatically appears on the worker’s profile. Upon completion, both parties publicly rate each other on a one-to-five scale.

3 Data and Stylized Facts

In this section, we outline our main data sources, including platform data, worker performance, and baseline surveys, and document key stylized facts that motivate our experiments.

3.1 Platform Data Collection

We collect three sources of platform data on data entry freelancers to document competition and wage bidding patterns: profile characteristics, job history, and job applications. We begin by describing our sample and key variables before presenting the findings.

Basic Freelancer Profile We scraped listings of all data entry workers located in Bangladesh, India, and Pakistan in May 2024. These three countries account for over 50% of the global freelancer population (Online Labour Observatory, 2025). The sample comprises 10,818 freelancers. For each, we collected basic profile information visible to employers in the platform’s search engine: number of completed jobs, total earnings, total hours worked, job success rate, preferred hourly rate, and agency association.

Job History To examine workers’ career trajectories on the platform, we obtained detailed job histories for all 4,887 data entry workers in Pakistan, who represent over 45% of freelancers across the three countries. For each completed job, we recorded its duration, contract type, earnings, and employer rating.

ensuring only serious applicants apply. While no public data exists on the fee amount, our primary data collection reveals fees between \$1.20 and \$1.65 per application. This fee requirement is unique to the studied platform, though workers on other platforms still incur time and effort costs when applying.

Job Applications Application data is not publicly available on the platform. As part of the freelancer experiment, we collected application records from 2,332 workers outside our experimental sample to assess competition and wage bidding patterns. These records include wage bids, cover letters, and basic profile characteristics such as number of completed jobs and total earnings.

3.2 Stylized Facts from Platform Data

Using these data, we document three patterns that together motivates the central puzzle: why don't novice freelancers compete on prices to gain entry?

Fact 1. *Novice freelancers face significant entry barriers.*

We begin by examining how many freelancers successfully secure work on the platform. Among all data entry workers, 44% have never completed a job, and 62% have completed two or fewer. This aligns with broader evidence: [International Labour Organization \(2021\)](#) reports that 27-77% of freelancers across five major platforms have never completed a job, while [Hilbert and Lu \(2020\)](#) estimate that only 10% of registered freelancers in Latin America have ever secured platform work, with jobs highly concentrated among established workers. These patterns suggest that breaking into the market is a significant hurdle for new entrants.

Fact 2. *Freelancers' job prospects improve with the number of completed jobs.*

Next, we ask whether early jobs shape subsequent outcomes. Using job history data, we examine freelancers' growth trajectories. Appendix Figure [A2](#) plots wait time until the next job and log earnings from the next job against the number of cumulative jobs completed. Those with more completed jobs experience shorter gaps between jobs and faster earnings growth. While these patterns likely reflect selection on worker ability, they illustrate why securing initial jobs may be particularly valuable for high-ability novices. This is consistent with existing findings that initial reputations benefit high-performing novices and harm low-performing ones in securing future work ([Pallais, 2014](#); [Fazio, Freund and Novella, 2025](#)).

Fact 3. *Competition is intense, yet wage bids vary little.*

Given the entry barriers and the returns to initial jobs, a natural question is whether novices compete on price. Application data reveals intense competition: each vacancy receives an average of 64 applications, 20% from experienced workers with at least three completed jobs. Despite this, there is little variation in wage bids. Appendix Figure [A3](#), 80% of applicants bid exactly at the posted budget, 15% bid below and 3% above. Among novices with fewer than three jobs, just 17% underbid.

This pattern is puzzling. If initial employment provides valuable reputation-building opportunities, rational novices should bid below experienced workers to increase their hiring

probability. We next explore potential explanations for the limited variation in wage bids.

3.3 Measuring Worker Performance and Beliefs

The limited wage variation could be explained by two potential reasons.⁶ First, in a competitive labor market where wages equal marginal products of labor, similar bids may reflect similar worker productivity. Second, workers may share beliefs about returns to wage undercutting that discourage bidding below posted budgets. Since neither novices' ability or beliefs can be observed from platform data, we collect primary data by hiring novices to complete a standardized task and baseline survey, then validating the performance measure through a follow-up task.

Main Sample Recruitment We posted a beginner-level data entry and online research job on the platform and hired freelancers over eight rounds between May and August 2024.⁷ Eligible applicants had to meet three criteria: (1) based in LMICs, (2) specialized in data entry, and (3) completed no more than one job. To capture active job seekers, we selected eligible applicants on a first-come, first-served basis until all slots were filled.⁸ The final sample comprised 481 novice freelancers.

Standardized Task Hired freelancers received detailed task instructions, which involved correcting existing entries against the original sources and adding new data from scanned PDF files into an Excel Spreadsheet within 24 hours.⁹ To incentivize worker effort and ensure our performance measure reflected true ability, we informed workers that the top 3 performers would receive additional bonuses.

Worker performance on the data entry task was evaluated based on three objective dimensions: accuracy, speed, and following instructions.¹⁰ Accuracy was measured as the percentage of correct entries, determined by comparing worker output against verified entries prepared by professional data entry specialists before the experiment. Speed measured the time between task release and final submission. Ability to follow instructions was scored

⁶An alternative explanation is that workers avoid bidding down wages due to threats of social sanction. This is unlikely in our setting. Prior literature documenting such effects focuses on village economies where workers are in close networks. In online marketplaces, freelancers operate independently and their actions are not observed by other workers. We confirmed this in our qualitative survey with freelancers: few of them expressed concerns that bidding low wages would be deemed unfair to others.

⁷We conducted the hiring in batches because bringing all workers onboard simultaneously would be logistically challenging and delay performance evaluation across the sample.

⁸To minimize attrition and reduce the risk of scams, we excluded freelancers who submitted unprofessional cover letters or proposed unreasonable wage bids.

⁹Although workers were allowed to use optical character recognition (OCR) or AI tools, manual work was necessary since the scanned PDFs were of poor quality and the spreadsheet required a specific format.

¹⁰To ensure these measures capture relevant dimensions of quality, we conducted extensive interviews with freelancers prior to the experiment and confirmed these three aspects were considered the most important performance indicators. In addition, this approach is similar to the performance measurement in [Pallais \(2014\)](#) and [Kala and Lyons \(2025\)](#).

from one to six, with each point reflecting adherence to a specific formatting requirement. To create comparable performance measures across workers, we ranked all freelancers within each round on each dimension and assigned quartile scores from 1 (lowest) to 4 (highest). The total performance score equals the sum of these three quartile scores, ranging from 3 to 12. Workers scoring above the median score within their round were classified as high-type, and others as low-type.¹¹

Baseline Survey Following the standardized task, workers completed an online survey, which collected demographics and job search behaviors to complement the platform data. Since the limited wage variation could stem from either similar ability or similar beliefs, we measured workers’ perceptions of both their own performance and the effectiveness of wage undercutting. To measure perceived performance, workers were incentivized to predict their performance quartile on each of the three evaluation dimensions, with cash bonuses for correct predictions. These predictions sum to a perceived performance score (ranging from 3 to 12) that can be compared to actual performance.

To measure perceived effectiveness of wage undercutting, we elicited worker beliefs about different approaches for novices to secure jobs, such as writing cover letters, paying the platform to boost their applications, and bidding below the job budget. Multiple strategies were included to mitigate concerns about experimenter demand effects. These beliefs can later be benchmarked against actual employer responses from the demand experiment to assess whether workers accurately understand the value of undercutting.

Follow-up Task To cross-validate our performance measure and compare novice versus veteran workers, we posted a second data entry job on the platform and extended 66 offers using stratified random sampling: 34 to in-sample and 32 to out-of-sample applicants. Among in-sample workers, we stratified by baseline performance and whether they bid below the posted budget to ensure a balanced sample for comparing subsequent performance across ability types and bidding strategies. Among out-of-sample applicants, we stratified by experience level, novice versus veteran, to allow for comparisons between new entrants and more established workers. Sixty-two applicants accepted the offer.

The task involved extracting data from financial reports into an Excel spreadsheet. Performance was measured using the same three dimensions as before, with one modification: because nearly half of workers followed instructions perfectly, we used a binary indicator for perfect compliance rather than quartiles. Therefore, the total performance score ranges from 3 to 10.

¹¹When freelancers had the same score, we broke ties by sequentially comparing their accuracy, ability to follow instructions, and speed.

3.4 Stylized Facts from Performance and Belief Data

Our main sample comprises 481 novice freelancers from 37 LMICs. Table 1 reports the sample characteristics measured in the baseline survey. Freelancers are roughly evenly split by gender and average 29 years old. Three features are particularly notable. First, the sample represents a large idle, skilled workforce: nearly half (48%) are not currently employed offline, yet 83% hold at least a bachelor’s degree. Second, workers struggle to break into the online market. Despite being registered for an average of 19 months, 87% have completed no jobs. Third, the entry barrier extends across platforms. While 44% have tried other freelancing platforms, only 18% have earned any money from them. Therefore, building reputation to access digital jobs is especially valuable for these novices, given their limited offline employment opportunities and persistent difficulty breaking into online markets.

Table 1: Freelancer Sample Characteristics

	Mean	Median	Min	Max
Female	0.52	1	0	1
Age	28.76	28	18	53
Bachelor Degree or Above	0.83	1	0	1
Employed Offline	0.52	1	0	1
Available as Full-Time Freelancer	0.36	0	0	1
Platform Tenure (Month)	19.65	12	0	278
Number of Applications/Month	9.80	7	1	73
Number of Completed Jobs	0.13	0	0	2
Present on Other Platforms	0.44	0	0	1
Has Earned from Other Platforms	0.18	0	0	1

Notes: This table reports sample characteristics for 481 freelancers from the baseline survey described in Section 3.3.

Next, we document patterns in novice freelancers’ ability and beliefs to motivate potential explanations for the limited wage variation.

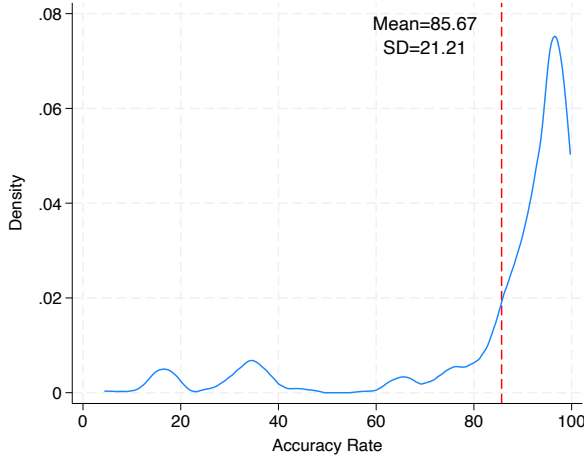
Fact 4. *Novices show substantial variation in task performance at baseline.*

We find heterogeneity in baseline task performance among novice workers along three dimensions: accuracy, speed, and ability to follow instructions. Figure 1 plots the distributions for each dimension and the total performance score, constructed by summing quartile rankings across the three measures. Accuracy and task completion time display wide dispersion, while instruction-following varies more moderately within the scale. The resulting total performance scores span the full range (3 to 12) and indicate substantial dispersion.

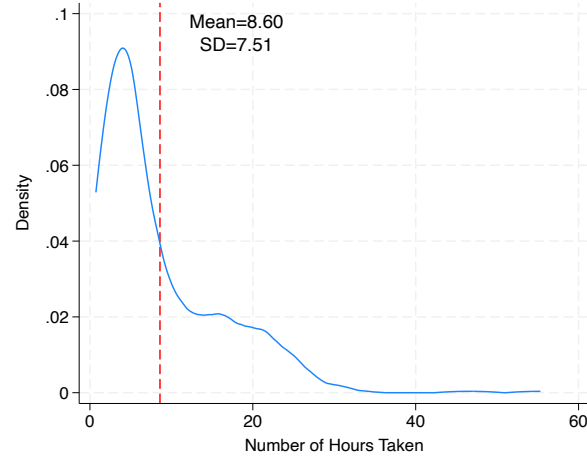
This variation in performance contradicts the hypothesis that limited wage variation reflects similar productivity among novices.

Figure 1: Distribution of Task Performance Among Novices

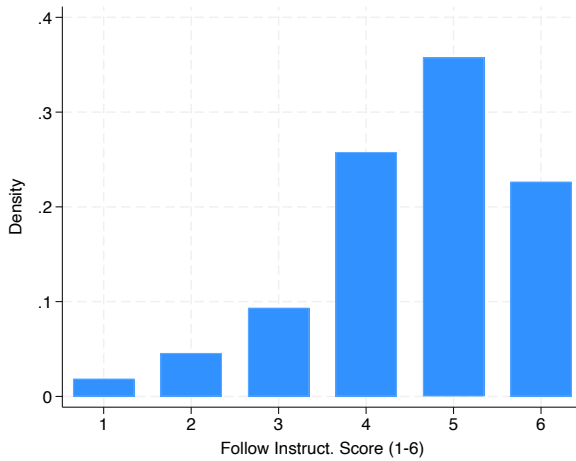
Panel A. Accuracy



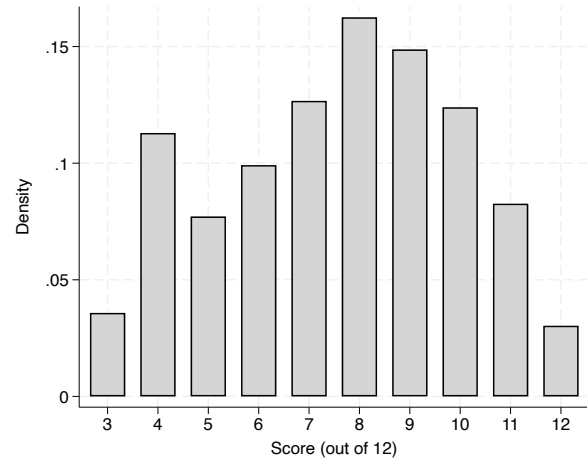
Panel B. Speed



Panel C. Follow Instruction



Panel D. Distribution of Performance Scores



Notes: This figure shows the distribution of the share of correct entries (accuracy), hours taken to complete the task (speed), the following instruction scores, and the total performance scores for the standardized data entry task.

Given heterogeneity in novices' baseline performance, we compare novices and veterans in the follow-up task to evaluate a central efficiency question: are novices less capable than veterans, potentially justifying their difficulty securing jobs? Appendix Figure A4 Panel A shows that novices outperform veterans on average. To ensure this pattern reflects genuine ability differences rather than noise, we examine whether performance among veterans increases with experience. Panel B confirms that among veterans, performance is positively correlated with completed jobs, as expected. Together, these patterns point to potential talent misallocation: capable novices face entry barriers despite outperforming many incumbents. That said, the sample is small ($n = 62$), so these findings should be interpreted as

suggestive rather than conclusive.

Fact 5. *Novices have limited knowledge of their own performance relative to others.*

When workers lack accurate information about their ability, wage bids are shaped by perceived rather than actual performance. Figure 2 Panel A compares novices’ true and perceived task performance and shows that most misperceive their performance quartile.¹² This suggests that limited wage variation is more likely driven by uncertainty about one’s own ability rather than by similarity in true productivity.

A potential concern is that the gap between actual and perceived performance reflects measurement error rather than true misbeliefs. To address this, we examine the correlation between novices’ performance in the two tasks and find a substantial positive relationship ($\rho = 0.53$), indicating that our performance measure captures a meaningful ability component rather than pure noise.

Fact 6. *Novices believe employers interpret low bids as signals of low quality.*

Lastly, we examine novices’ belief about how employers respond to low wage offers. As shown in Figure 2 Panel B, 44% of freelancers agreed with the statement in the baseline survey: “If a new freelancer offers a lower price, the client may think that their work is low quality.” This concern is consistent across worker types by baseline performance, suggesting that perceived employer responses may deter novices from underbidding to avoid sending a negative quality signal.

Taken together, these findings point to two mechanisms behind limited wage undercutting among novices: uncertainty about one’s own ability and beliefs about how employers interpret low wages. These descriptive patterns motivate our experimental design, which causally tests these mechanisms on both the demand and supply sides.

4 Demand Experiment

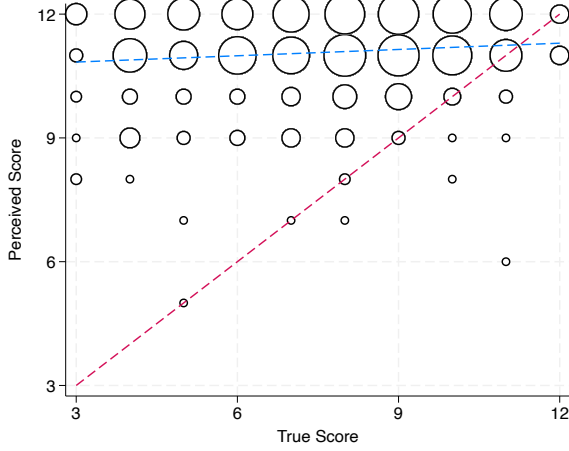
4.1 Experimental Design

Given novices’ beliefs about employer responses to low wages, we first test whether employers in fact interpret low bids as signals of low quality and whether undercutting improves novices’ hiring prospects. Answering this question with observational data is challenging for two reasons. First, employers’ perceptions of worker quality cannot be directly observed. Second, hiring decisions depend on both wage bids and other application characteristics, making it difficult to isolate the causal effect of undercutting on employer responses.

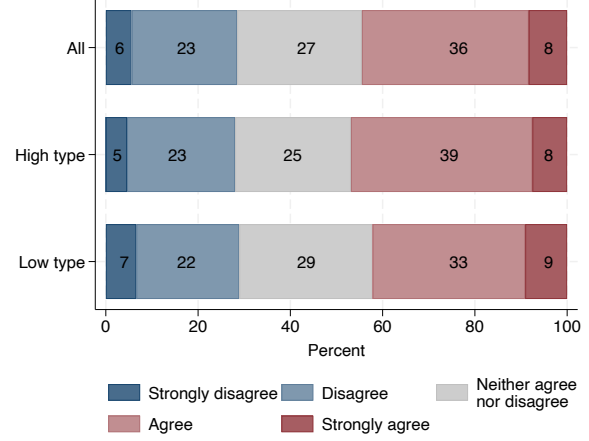
¹²Appendix Figure A5 presents the comparison for each dimension and finds similar patterns.

Figure 2: Novices' Baseline Beliefs

Panel A. Actual versus Perceived Performance



Panel B. Low Bids = Low Quality to Employers



Notes: The figure reports novices baseline beliefs about their own performance and the effectiveness of wage undercutting. Panel A plots true versus perceived standardized task scores; the red dashed line is the 45-degree line and the blue line shows the best linear fit. Marker size reflects the number of observations. Panel B shows responses to the statement: *If a new freelancer offers a lower price, the client may think their work is low quality.* Bars correspond to the full sample, above-median (high-type), and below-median (low-type) novices.

To overcome these challenges, we leverage the following identification strategy. If employers treat low wages from novices as negative quality signals, then lowering a novice's bid reduces hiring costs but also lowers perceived quality, so the effect on demand is ambiguous. In contrast, for veterans with public reputations for quality, a lower bid primarily reduces cost and should increase demand. This yields a testable prediction: holding all other application attributes constant, lowering wages should increase employer demand less for novices than for veterans.

We test this hypothesis using an audit-style experiment on the platform, in which we create comparable novice and veteran profiles and compare employer responses to different wage bids from these profiles. The experimental estimates then allow us to quantify private returns to wage undercutting for novice workers through a back-of-the-envelope calculation.

4.1.1 Implementation

Freelancer Profile Creation To isolate the causal effect of wage bids on employer demand, we need to hold all profile attributes constant except the experience level. We therefore created four freelancer profiles located in Pakistan and specializing in data entry and virtual assistance.¹³ Profile characteristics, including preferred hourly wages, self-reported skills, and language proficiency, were randomly assigned to reflect typical profiles in our descriptive

¹³Pakistan has the worlds third-largest online freelancer population and the highest share of data entry freelancers in our platform data.

data. Two profiles were randomly designated as novices with no job history. The remaining two were assigned veteran status, for which we built job histories between February and April 2025. Both veteran profiles were given identical numbers of completed jobs, job-specific ratings, and overall success rates to hold observable quality signals constant.

Job Listing and Application We compiled daily job listings from the platform between May and August 2025. Eligible jobs were data entry or virtual assistant positions with a fixed budget of at least \$7 USD,¹⁴ requiring entry- or intermediate-level skills. We submitted four applications per eligible job, each with a wage bid and a unique cover letter.¹⁵ Within each experience level, one application was randomly assigned to undercut by bidding 80% of the posted job budget, while the other bid the full budget. We chose the 20% threshold because it corresponds to the most common range of wage undercutting observed in our job application data, as shown in Appendix Figure A6. This design generated four application types per job: novice bidding 80% of the budget, novice bidding the full budget, veteran bidding 80% of the budget, and veteran bidding the full budget. Appendix Figure A1 illustrates how these applications appeared on the employer dashboard.

We tracked two measures of employer responses using unique platform features. First, *employer view* captures when employers click on an application to review the cover letter and freelancer profile, which automatically generates a platform notification. Second, when employers contact applicants to request information or initiate an offer, the platform automatically sends an email notification to the applicant, which we record as *callback*. We declined all callback messages to avoid interfering with employers’ hiring decisions and to maintain comparable job histories across experimental profiles.

4.1.2 Data

We submitted 2,972 applications to 743 job vacancies on the freelancing platform. In some cases, an application deviated from its pre-assigned wage or could not be submitted because the vacancy had already been closed or removed. We restricted the analysis sample to vacancies where all four applications were submitted with correct wages, yielding 2,812 applications to 703 vacancies.

Job and Employer Characteristics For each vacancy, we recorded basic job information, such as job title and budget, at the application time. Two weeks later, we tracked hiring outcomes and employer characteristics from the original job post, including the number of

¹⁴The platform sets a minimum price of \$5 USD for fixed-budget projects. We applied a \$7 USD threshold to ensure all submitted bids complied with this rule.

¹⁵We construct a distinct cover letter template for each profile, which is used across all applications submitted by that profile. To maintain authenticity and reduce the risk of detection, we tailor each template to the specific job posting. In the main analysis, we control for profile and job fixed effects to isolate the effects of cover letters from the treatment effects.

freelancers hired, employer location, tenure on the platform, number of past jobs posted, and employer ratings from previously hired freelancers. Appendix Table A1 summarize these characteristics.

Application Characteristics and Outcomes For each application, we recorded the experience level (novice or veteran), wage bids, cover letter, submission time, employer view, and callback. Appendix Table A2 summarizes the categories of callback messages.

4.2 Empirical Strategy

We estimate the causal effect of low wage bids on the likelihood of receiving employer responses by worker experience. The main regression specification is

$$(1) \quad Y_{ij} = \alpha + \beta_1 \text{Novice}_i + \beta_2 \text{LowBid}_{ij} + \beta_3 \text{Novice}_i \times \text{LowBid}_{ij} + \mu_j + \varepsilon_{ij}$$

where Y_{ij} is the response to the application from profile i to job j , Novice_i is an indicator equal to one if profile i has zero completed jobs, LowBid_{ij} equals to one if the wage bid is 20% below the job budget, and μ_j is the job fixed effects to improve precision of our estimates. Standard errors are clustered at the job level. By comparing employer responses to wage bids across novice and veteran profiles, we map testable predictions to the signs of coefficients.

Prediction 1a. *If employers interpret lower wages as negative quality signals in the absence of other credible information, labor demand will be more responsive to low wage bids from veteran workers than from novices: $\beta_2 > 0$, $\beta_3 < 0$.*

Prediction 1b. *If low wages are not perceived as negative signals, undercutting will increase hiring chances for both novices and veterans, equally or more for novices: $\beta_2 > 0$, $\beta_3 \geq 0$.*

If Prediction 1a holds, low wage bids will have competing effects on demand through reduced hiring costs and lower perceived quality. When quality concerns dominate, low bids harm novices' chance of getting hired: $\beta_2 + \beta_3 < 0$. When hiring costs dominate, novices can improve their hiring chances by bidding low wages: $\beta_2 + \beta_3 > 0$, $\beta_3 < 0$.

4.3 Impact of Lowering Bids on Employer View and Callback

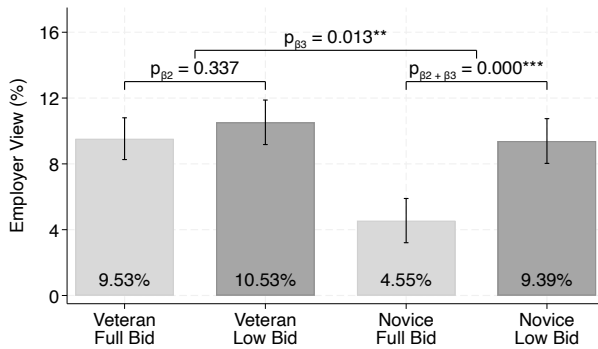
We find that low wage bids increase employer views and callbacks for both novices and veterans, with larger effects for novices. Figure 3 presents estimates from specification 1. Results are robust to alternative specifications and sample restrictions as shown in Appendix Table A3.

Panel A shows effects on employer views. At the posted budget, veterans are more than twice as likely as novices to have their applications viewed. Undercutting to 80% of the budget

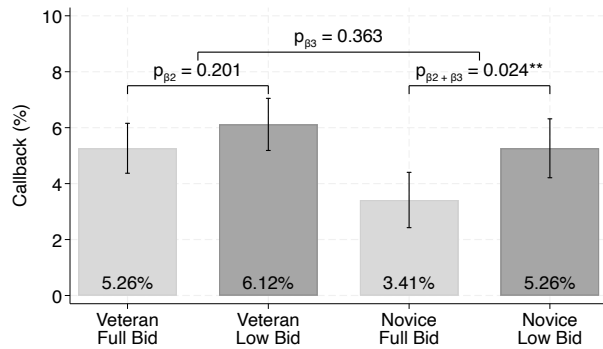
more than doubles views for novices (from 4.55 to 9.39 percentage points), while increasing views only modestly for veterans (from 9.53 to 10.53 percentage points). Statistical tests on β_3 and $\beta_2 + \beta_3$ reject the hypothesis that employers interpret low bids from novices as negative quality signals ($p < 0.001$ and $p = 0.013$). Panel B reports similar patterns for callbacks. Low bids raise callback rates by 1.85 percentage points for novices and 0.86 percentage points for veterans, corresponding to increases of 54% and 16% relative to baseline. While the interaction is not statistically significant, undercutting significantly improves novices' callback probability as $\beta_2 + \beta_3$ is significantly different from zero ($p = 0.024$).

Figure 3: Employer Responses to Low Bids

Panel A. Employer Views



Panel B. Callback



Notes: The figures show results from regression 1 for employer views and callback. Each bar represents the sum of the control mean (veteran bidding full budget) and the relevant regression coefficients. We report the 95% confidence intervals based on the estimated standard errors of the linear combinations of the regression coefficients and p-values from Wald tests.

Appendix Figure A7 presents the estimates as “demand curves”, showing that employer views and callbacks decline with wage bids for novices and are more elastic than for veterans. These findings are consistent with Prediction 1b: employers do not treat low wage bids from novices as negative quality signals. Instead, lowering wages improves novices' hiring prospects, contrary to their baseline beliefs.

While employers are, on average, more likely to consider novice workers when they bid low wages, we observe meaningful heterogeneity across employer characteristics, as shown in Appendix Table A4.

Two patterns stand out. First, less experienced employers exhibit more elastic demand for novices than more experienced ones. Measured by platform tenure and number of jobs posted, less experienced employers are substantially less likely to respond to novices who bid at the job budget than to veterans, but are much more responsive to those offering low wage bids. In contrast, experienced employers show a smaller differential response to undercutting. One explanation is that experienced employers may already have established relationships with freelancers, making them less sensitive to low bids from unknown appli-

cants. By contrast, less experienced employers, who lack a stable pool of past collaborators, may rely more heavily on wage bids as a screening device when evaluating novice applicants. Second, higher-rated employers are significantly more likely to respond positively to low bids from novices than lower-rated employers. This pattern alleviates concerns about adverse selection, whereby undercutting might disproportionately attract exploitative or low-quality employers.

Taken together, these results suggest that lowering wage bids improves novices' hiring prospects, particularly with newer employers on the platform, without steering them toward lower-quality matches.

4.4 Returns to Lowering Wage Bids

Although lowering wage bids is an effective entry strategy, novices may still avoid undercutting if the future benefits from reputation building do not outweigh the immediate loss in earnings. To assess this concern, we conduct a back-of-the-envelope calculation comparing novices' job arrival and net earnings under two strategies: always bidding at the full job budget and consistently undercutting the budget by 20 percent. This exercise allows us to quantify the private returns to offering low wages.

We begin by modeling job arrival. Suppose a novice submits a applications, pays an application cost c_a per job, and receives an offer with probability p , which we approximate using callback rates from the demand experiment. Because jobs in this setting are usually short in duration, we assume novices are not capacity constrained and therefore always accept job offers. The expected number of jobs is then $n(a) = a \times p$.

Figure 4 Panel A plots this relationship between applications and accumulated jobs under the two bidding strategies. Undercutting increases p , thereby accelerating job arrival. For example, to secure the first four jobs, novices who undercut by 20 percent need to submit 76 applications whereas those bidding at the full budget require 118 applications. Given that novices in our baseline sample submit approximately 10 applications per month, undercutting shortens the time to four jobs by about four months.

Next, we compare novices' net earnings on the platform. Let the net return from the first job equal the wage bid minus the effort cost: $\pi_1 - c_e$. As workers accumulate experience, earnings grow at rate $\rho(n)$, reflecting the returns to reputation documented in the job history data. We calibrate these parameters using estimates from the demand experiment, administrative job histories, and the baseline survey; Appendix Table A5 reports all parameter values and data sources. Novices total net earnings as a function of applications can be written as:

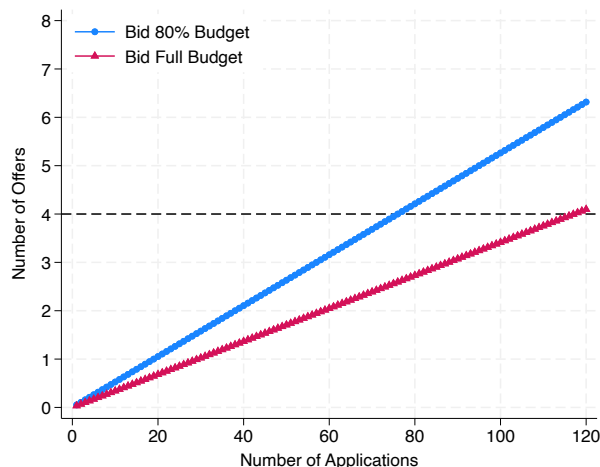
$$\Pi(a) = a \times (p \times (\pi_1 - c_e) \times \rho(n) - c_a)$$

Figure 4 Panel B plots net earnings under the two strategies. Undercutting dominates bidding at the full budget: novices earn more because they obtain jobs faster, experience earlier earnings growth, and incur lower application costs.

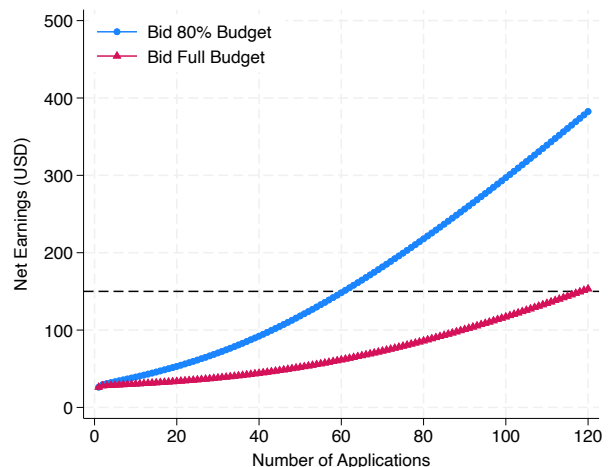
The magnitudes are economically meaningful. To earn the first \$150 USD, novices need 61 applications when consistently undercutting, compared to 119 applications when bidding at the full budget—a difference equivalent to six months of search. Over one year (120 applications), workers who never undercut earn about \$153 USD, whereas those who always undercut earn \$382 USD, nearly 2.5 times as much. The \$229 USD gap corresponds to roughly 20 percent of annual income in low-income countries.

Figure 4: Expected Number of Jobs and Net Earnings

Panel A. Expected Number of Jobs



Panel B. Expected Net Earnings



Notes: This figure presents back-of-the-envelope calculations comparing two bidding strategies for novice freelancers: bidding at 80 percent of the job budget and bidding at the full budget. Panel A shows the expected number of jobs as a function of applications. Panel B shows cumulative net earnings (USD), incorporating wage bids, effort costs, earnings growth from reputation, and application costs. Parameter values and data sources are reported in Appendix Table A5. Dashed lines mark four jobs (Panel A) and \$150 (Panel B) for reference..

These estimates are illustrative and may not fully capture actual returns for several reasons. First, we only measure effects of undercutting by 20 percent, while actual undercutting amounts vary widely (Appendix Figure A6). Second, we assume one-to-one conversion from callbacks to job offers, which likely provides an upper bound on platform earnings since we do not directly observe conversion rates. Third, we compare pure strategies rather than the mixed strategies workers might employ in practice. Workers may optimize by undercutting selectively based on job characteristics or market conditions.

Finally, our calculation captures private returns and abstracts from general equilibrium effects. A natural concern is that widespread undercutting practices by novices could trigger a race to the bottom in wages. However, this need not be the case. First, Cui (2025) documents

market expansion in a similar online setting, where low initial bids by novices substantially increased total matches on the platform. Second, we find evidence of slack demand: only 54 percent of jobs in our demand experiment were filled, as shown in Appendix Table A1. This suggests scope for price competition to move the extensive margin by increasing the likelihood of successful job matches, rather than merely reallocating existing matches across workers.

Despite these limitations, the results demonstrate that wage undercutting offers substantial economic gains for workers in developing countries seeking to overcome entry barriers in online platforms. This raises a key question: given these returns, why don't novice workers offer lower wages to break into the market?

5 Freelancer Experiment: Design

The freelancer experiment examines why few novice workers offer low wages to overcome entry barriers. The baseline survey identifies two potential frictions: novices' limited information about their own abilities and inaccurate belief that employers interpret low wages as low quality signals. We test how correcting these frictions affects novices' bidding decisions.

5.1 Conceptual Framework

Our experiment is motivated by a standard reputation model (Tirole, 1988) with supply-side belief frictions. We outline the main intuition here and present the full model in Appendix B. In this labor market, employers live for one period, while workers live for two: they enter as novices and, if hired, become veterans with publicly observable reviews. Employers post jobs with a stated budget, and workers apply by submitting wage bids at or below this budget.¹⁶ Novices differ in innate ability (high or low), which is known to themselves but not to employers ex ante. Employers can infer veterans' ability from reviews. For novices, employers form beliefs based on observed wage bids and may incur a cost to screen applicants. Workers choose wage bids to maximize expected lifetime earnings, and employers hire the applicant who generates the highest expected return.

In the baseline model, high-ability novices offer lower wages initially to compensate employers for hiring them and build reputation for future jobs, while low-ability novices maximize current earnings since they won't be hired again once their type is revealed. This strategy can be sustained when the future returns to reputation exceed the short run cost of lower wages for high-ability novices. This condition is likely to hold in our setting, given the high returns to reputation and relatively low outside options. In equilibrium, low wage bids serve as credible signals of high ability, allowing employers to infer quality without

¹⁶This is empirically motivated: only 3% of applicants in our job application data bid above the budget.

incurring screening costs. This yields a separating equilibrium in which only high-ability novices undercut, are more likely to be hired, and deliver higher-quality output.¹⁷

Two supply-side belief frictions can cause the market to deviate from the separating equilibrium. First, if novices are uncertain about their own ability, wage bids reflect expected rather than actual ability, weakening the signaling value of low bids. Employers must then rely more on costly screening, making novices more expensive to hire. As a result, fewer high-ability novices are hired, and match quality worsens. Second, if novices believe that employers interpret low wages as signals of low quality and place greater weight on these concerns in hiring, they pool at the posted budget to avoid being perceived as low quality. This eliminates the informational content of wage bids. Employers must then incur additional screening costs to distinguish among novices and, with identical bids, cannot update their beliefs about the relationship between wages and quality. Both frictions impede high-ability novices from signaling their quality, slowing talent discovery and limiting their ability to overcome entry barriers.

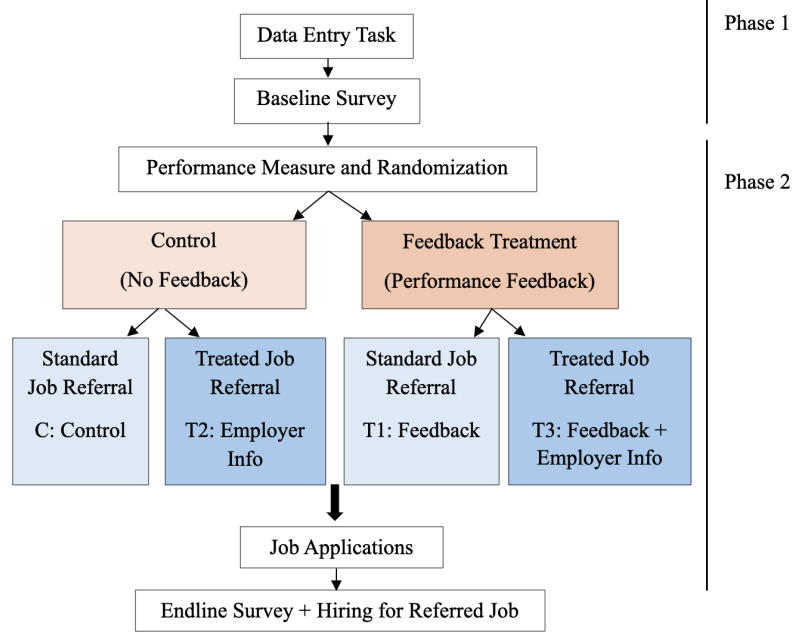
Our experimental design shuts down each of the frictions by exogeneously providing novices with information about their own ability and/or about employer responses to low wage bids. The framework yields three predictions. First, in the absence of misperceptions about employer responses, providing novices with information about their own ability induces separation in wage bids, driven by high-ability novices. Second, correcting beliefs about employer responses increases undercutting among novices. Third, when both frictions are alleviated, the increase in undercutting should again be concentrated among high-ability novices. Next, we describe the experimental design.

5.2 Timeline and Randomization

Our experimental design consists of two phases. In the first phase, we hired novice freelancers for a standardized data entry task to measure their baseline performance and beliefs. In the second phase, we randomly provided these workers with information about their performance and employer responses, and tracked their subsequent job applications as main outcomes. Figure 5 presents the experimental design.

¹⁷The baseline model rests on assumptions consistent with recent empirical evidence. First, the assumption that employers live for only one period rules out incentives on the demand side to invest in training. This is realistic in our setting: the average hourly contract length among novices in our data is 6.4 hours, and most freelancers work with unique employers (Pallais, 2014). Second, we assume that low-ability workers do not improve their productivity through accumulating platform experience because once their abilities are revealed, they will not acquire future jobs on the platform. Fazio, Freund and Novella (2025) find suggestive evidence that low ratings on initial jobs reduce workers' chances of securing further work. Thus, online platforms provide limited scope for skill accumulation after a bad initial signal. As a result, only high-type novices have strong incentives to offer low wages in their first period in order to establish a reputation for future employment.

Figure 5: Freelancer Experiment Timeline



Notes: This figure shows the freelancer experimental design. This design was repeated for 8 rounds and each round took place over 2 weeks.

Section 3.3 describes the first phase in detail, including the sample recruitment, the standardized task and performance evaluation, and the baseline survey. Following the performance evaluation, we cross-randomized two treatments at the individual level: the *feedback treatment* and the *employer information treatment*. The randomization was stratified by whether workers performed above the median (high-type) or below the median (low-type).

Feedback Treatment To address workers’ limited knowledge about their performance, we randomly assigned half of the sample to receive private feedback about their baseline performance. Workers were told that we evaluated their work based on accuracy, speed, and following instructions relative to other novice freelancers hired for this job. For each dimension, they received a score from 1 to 4 stars, representing their quartile rank. Below is an example of the feedback.

Hi! We reviewed your data entry work and compared it with other new freelancers we hired. We looked at three things: accuracy of the entry, the time taken to complete the task, and how well you followed instructions. Your results:
Accuracy: ★★ - below average, but not in the bottom 25%
Speed: ★ - bottom 25%
Following instructions: ★★★ - above average, but not in the top 25%
Please note that this will not affect our review for you. We ONLY share this feedback with you and hope this helps you grow and succeed on this platform.

A few days after the feedback treatment, we posted a data entry job from another employer accounts on the platform. Appendix Figure A8 shows that the job budgets were comparable to typical data entry jobs as measured in the demand experiment and well above the \$5 minimum for fixed-payment projects. We then shared the job link with all workers in the experimental sample through our original account.¹⁸ The standard referral message was as follows:

Hi! A former colleague of ours is hiring for a data entry job and asked us to help spread the word widely. They mentioned they are open to working with new freelancers. If you're interested, feel free to apply directly! Since we are not involved in the hiring process, we won't be able to make individual recommendations.

Employer Information Treatment To address workers' belief that employers treat low wage bids as negative signals, we cross-randomized half the sample to receive additional information about the employer. The treated referral message was as follows:

*Hi! A former colleague of us is hiring for a data entry job and asked us to help spread the word widely. They mentioned they are open to working with new freelancers and **won't judge quality of new freelancers based on their proposed prices.** If you're interested, feel free to apply directly! Since we are not involved in the hiring process, we won't be able to make individual recommendations.*

This referral design helps establish the credibility of the employer information for novices while preserving their incentives to attract employers unfamiliar with their abilities. One concern is that the treated message might eliminate workers' incentives to signal *high* quality by bidding low wages. However, baseline survey data suggest this is unlikely: fewer than 4 percent of workers believed that employers would interpret low wage bids from novices as high-quality signals. The neutral tone of the message also minimizes experimenter demand effects or suspicion about the job, which we later confirm using responses from the endline survey.

After the treatment, we tracked applications received within two days, as most freelancers responded within 24 hours. Application data, including wage bids and cover letters, were scraped from the employer dashboard and include both in-sample and out-of-sample applicants. Our main outcomes are whether novices in the experimental sample submit an application and the wage bids they propose.

Endline Survey We administered the endline survey to in-sample novices after the two-day application window. The survey measures workers' perceptions of the referred job and

¹⁸Within each round, we posted one job from two different employer accounts to avoid over-crowding one application pool. Workers were randomly assigned to one job opportunity.

re-elicits their beliefs about their performance relative to other novices. These data allow us to assess treatment validity, that is, whether the intervention shifted novices’ beliefs as intended. For those who did not apply, we also asked about their reasons for not submitting an application. After completing the survey, we closed the initial contract and left a public but uninformative review for all in-sample novices.

Follow-up Task We randomly offered the follow-up job to a small subset of applicants. Section 3.3 describes the task content and hiring procedure in detail. We then tracked offer acceptance and worker performance to examine treatment effects on these downstream outcomes.

5.3 Experiment Validity

We now report several checks that support the validity of the experimental implementation. First, we verify that baseline freelancer characteristics are balanced across treatment and control groups. These characteristics were collected through the baseline survey, including demographics, work status, platform experience, and performance measures. We supplement these with public profile characteristics scraped from the platform, including number of jobs completed, preferred hourly wages, location, English proficiency, professional certificates, and other background details. Appendix Table A6 reports summary statistics and compares freelancers along 20 baseline and profile characteristics. Of the 80 tests, two are statistically significant at the 5% level and three are statistically significant at 10% level, as expected under random chance.

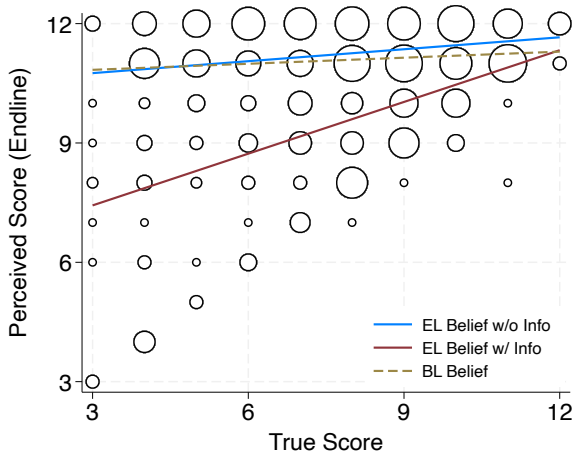
Second, attrition in the endline survey is minimal. Only 1.2 percent of the experimental sample did not complete the endline survey because their accounts were blocked by the platform and we were not able to reach them.

Finally, we provide evidence of treatment validity by showing that workers correctly interpreted the information provided. Figure 6 shows the treatments worked as intended. Panel A compares novices’ perceived performance scores at the endline with their actual scores by treatment status. Workers who did not receive feedback show little change in their assessments: their endline beliefs closely align with the baseline ones, which indicates little updating. In contrast, workers who received private performance feedback revise their beliefs substantially, with perceived scores moving closer to the true performance level represented by the 45-degree line. In the treated groups, this updating occurs across the performance distribution rather than being driven by high- or low-type novices alone. Among novices with lower baseline performance, belief revisions are larger but the updated beliefs are still away from the truth. Among higher-performing novices, belief adjustments are smaller but their updated beliefs align more closely with their true performance. We also show in Appendix Figure A9 that similar patterns emerge when examining the three dimensions of perfor-

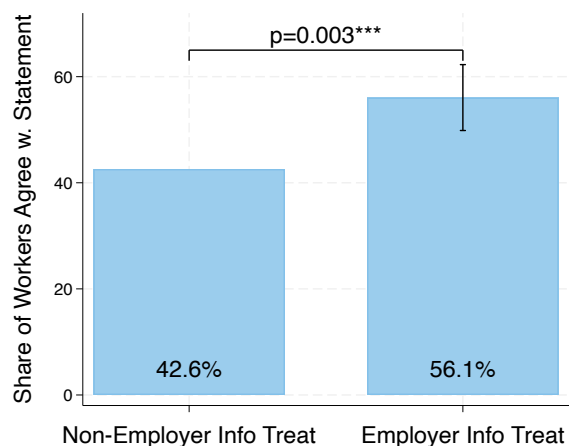
mance separately. Overall, these results confirm that the feedback treatment successfully shifts workers' beliefs about their performance relative to other novices.

Figure 6: Treatment Effects on Worker Beliefs

Panel A. Beliefs about Own Performance



Panel B. Employers Don't Infer Quality from Bids



Notes: This figure reports the treatment effects on worker endline beliefs about own performance and employer responses. Panel A plots workers' actual performance scores against their perceived scores measured at the endline. The blue line represents the best linear fit for workers without performance feedback and the red line for workers with performance feedback. The dotted line shows the best linear fit of baseline perceived performance against true performance. Panel B reports the share of workers who "agree" or "strongly agree" with the statement: "*The employer (of the referred job) will not judge my quality based on my wage bids.*" We report the 95% confidence intervals and p-values from the Wald test in Panel B.

Panel B examines whether the employer information treatment alleviated novices' concerns that employers interpret low wage bids as signals of poor quality. In the endline survey, workers were asked to indicate whether they agree with several statements about the referred job. One statement reads: "*The employer (of the referred job) will not judge my quality based on my wage bids.*" We find that in the treated groups, the proportion of workers agreeing with the statement rose by 13.5 percentage points, representing a 32 percent increase from the non-treated mean. This difference is statistically significant ($p = 0.003$), indicating that the employer information treatment meaningfully shifted workers' beliefs about how employers interpret wage bids. Appendix Figure A10 further reports novices' baseline and endline beliefs about employer responses to low wage bids and shows that the treatment increases the share of workers holding the correct belief, while baseline beliefs remain similar across treatment groups. This provides additional evidence that the feedback treatment shifted beliefs as intended rather than reflecting pre-existing differences between treated and non-treated workers.

Importantly, the treatment does not appear to affect workers' perceptions of the credibility or competitiveness of the job itself. Appendix Figure A11 shows that 83 percent of novices view the referred job posting as trustworthy and 85 percent believe competition for the job

is strong, with no significant differences across treatment groups.

6 Freelancer Experiment: Results

In this section, we outline the empirical strategy and present the experimental results on job applications, wage bids, offer acceptance, and performance to test the predictions from the conceptual framework. We then examine the role of baseline beliefs to shed light on the underlying mechanisms and evaluate alternative hypotheses.

6.1 Empirical Strategy

We estimate the following specification to measure the treatment effects on novice freelancers likelihood of wage undercutting:

$$(2) \quad \text{Undercut}_i = \alpha + \beta_1 \text{Feedback}_i + \beta_2 \text{EmployerInfo}_i + \beta_3 (\text{Feedback} + \text{EmployerInfo})_i + \eta_r + \varepsilon_i$$

where Undercut_i is an indicator equal to one if novice i submits a wage bid below the posted budget. Feedback_i , EmployerInfo_i , and $(\text{Feedback} + \text{EmployerInfo})_i$ are treatment indicators. η_r denotes experiment-round fixed effects, and ε_i is an error term; we report Huber-White robust standard errors. The signs of the coefficients β 's capture the average treatment effects on novice i 's decision to undercut and correspond to the following testable predictions:

Prediction 2a. *If novice workers hold inaccurate beliefs about employer responses to low wage bids and have limited information about their own ability, all three treatments will affect their likelihood of wage undercutting: $\beta_m \neq 0$ for all $m \in 1, 2, 3$.*

Prediction 2b. *If novice workers believe that employers interpret lower wage bids as negative signals of quality, alleviating this concern (*EmployerInfo*) will increase their likelihood of wage undercutting: $\beta_2 > 0$.*

The signs of β_1 and β_3 depend on novices' prior beliefs about their own ability. In the conceptual framework, novices whose beliefs adjust downward are less likely to undercut as a form of reputation investment, whereas those whose beliefs adjust upward are more likely to do so. Therefore, the signs are ambiguous *ex ante*.

To capture differential treatment effects by novice ability, we include interactions of treatment

dummies with novices' baseline performance to the specification:

$$(3) \quad \text{Undercut}_i = \alpha + \theta_1 \text{Feedback}_i + \theta_2 \text{EmployerInfo}_i + \theta_3 (\text{Feedback} + \text{EmployerInfo})_i \\ + \delta \text{High}_i + \gamma_1 \text{Feedback}_i \times \text{High}_i + \gamma_2 \text{EmployerInfo}_i \times \text{High}_i \\ + \gamma_3 (\text{Feedback} + \text{EmployerInfo})_i \times \text{High}_i + \eta_r + \varepsilon_i$$

where High_i indicates worker i 's baseline performance was above the median. In this specification, δ captures differences in the likelihood to undercut between high- and low-type workers in the control group, θ 's capture the average treatment effects for low-type workers, and γ 's capture the differential treatment effects for high-type workers. We can map signs of these coefficients to the following testable predictions on wage bids by novice ability.

Prediction 3a. *If novice workers have limited information about their own ability, wage bids will not vary by worker ability: $\delta = 0, \gamma_2 = 0$.*

Prediction 3b. *If novice workers hold accurate beliefs about employer responses to low wage bids, providing them information about their ability (Feedback) will induce separation in wage bids, with high-type workers more likely to undercut: $\gamma_1 > 0$.*

Prediction 3c. *If novice workers hold inaccurate beliefs about employer responses to low wage bids, providing them information about both their ability and employer response (Feedback + EmployerInfo) will induce high-type workers to undercut: $\gamma_3 > 0$.*

6.2 Treatment Effects on Worker Outcomes

6.2.1 Job Application and Wage Bids

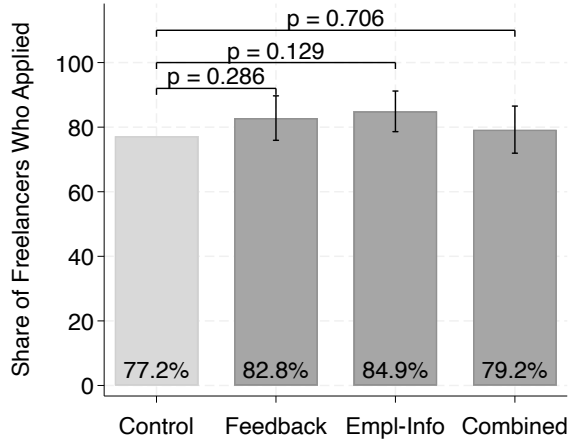
We find high compliance among novice freelancers in applying for the follow-up job across all four treatment groups. Figure 7, Panel A shows that over 80% of freelancers applied, with no evidence of differential attrition across treatment groups. Endline survey evidence is consistent with this pattern: among non-applicants, the most commonly cited reasons for not applying were a perceived low likelihood of selection due to competition (28%) and insufficient funds to cover the application fee (25%). This suggests that the treatments do not affect participation decisions and that subsequent results are not driven by selection into the applicant pool.

We next examine the main outcome: wage bids by novices. Figure 7, Panel B shows the share of freelancers bidding below the job budget across treatment groups.¹⁹ Consistent with our descriptive evidence, undercutting is rare in the control group: only 5.5 percent

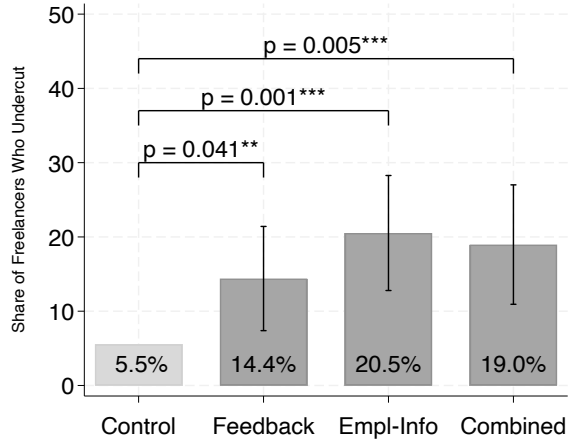
¹⁹Only one worker in our sample proposed a wage above the job budget.

Figure 7: Treatment Effects on Job Application and Wage Bids

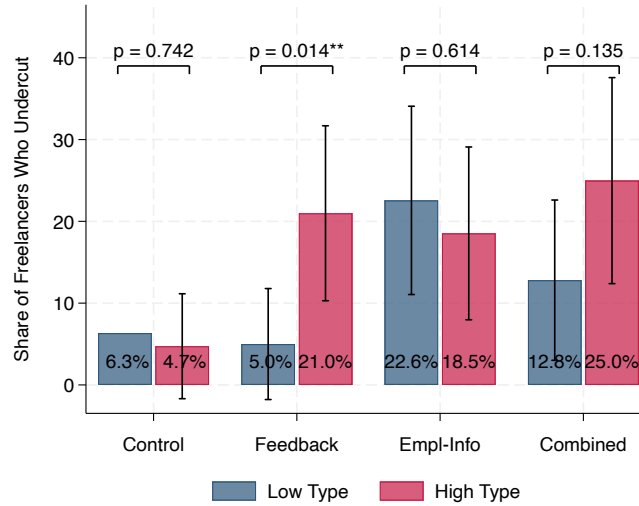
Panel A. Application Decisions



Panel B. Bids Below Job Budget



Panel C. Bids Below Job Budget by Worker Type



Notes: The figures show treatment effects on job application and wage bids. Panel A shows the treatment effects on whether workers apply to the referred job and Panel B shows effects on whether workers bid below the job budget conditional on applying based on regression 2. Panel C show heterogeneous treatment effects on wage undercutting by worker baseline performance (above or below median) from regression 3. Each bar represents the sum of the control group mean and the relevant regression coefficients. We report the 95% confidence intervals based on the estimated standard errors of the linear combination of regression coefficients and p-values from Wald tests. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

of freelancers bid below the budget.²⁰ All three treatments significantly shifted the share of novices who undercut, consistent with Prediction 2a. Relative to the control group, freelancers are 8.9 percentage points more likely to bid below the budget in the *Feedback* group ($p = 0.041$), 15 percentage points more in the *Employer-Info* group ($p = 0.001$), and

²⁰This rate is lower than in the descriptive data, likely because experimental participants received a job referral, which may have provided a perceived advantage over other applicants.

13.5 percentage points more in the *Feedback+Employer-Info* group ($p = 0.005$). These effects indicate that inaccurate beliefs about employer responses and own ability at baseline discourage novices from bidding low, and that correcting these beliefs substantially influences wage-setting behavior. The results also support Prediction 2b: providing information about employer responses increases novices’ willingness to undercut. Since the effects of the *Feedback* and *Feedback+Employer-Info* treatments are ambiguous ex-ante, we explore heterogeneity in baseline beliefs to explain the observed patterns in Section 6.3.

The standard reputation model predicts that high- and low-ability novices differ in their propensity to adopt “introductory pricing”. Do they respond differently to the treatments? Figure 7, Panel C shows the main results separately for novices who perform above (high-type) or below (low-type) the median in the baseline task. In the absence of performance feedback, the two types are equally likely to bid at the job budget in both the control group ($p = 0.742$) and the *Employer-Info* group ($p = 0.614$). Once performance feedback is provided, treatment effects are driven by high-type workers. High-type novices are four times more likely to bid below the budget than low-type novices in the *Feedback* group ($p = 0.014$) and twice as likely in the *Feedback+Employer-Info* group ($p = 0.135$). This pattern is consistent with the model, in which high-type novices have greater returns to securing initial jobs to build reputation and are therefore more likely to undercut initially.

Taken together, these results identify supply-side belief frictions that deter novices from using low wages as an entry strategy. In particular, uncertainty about own ability prevents the separating equilibrium in wage bids predicted by standard reputation models. Providing information about employer responses encourages both types to undercut, leading to pooling in wage bids. In contrast, providing information about own ability induces high-type novices to undercut. This separation improves match quality: high-type workers are more likely to be hired, deliver higher-quality output, and build reputations, thereby accelerating talent discovery in the market.

6.2.2 Other Application Features and Job Performance

We have shown that providing information about own ability and employer response increases novices’ propensity to bid below the posted budget, particularly among the high-ability workers. A potential concern is that these effects are driven by changes in other aspects of the application. We examine two alternative explanations: effort in cover letter writing and the timing of wage bargaining. First, since wage bids and cover letters are the two main choices applicants make in the application, treatments may shift effort from bidding to cover letter writing. Using cover letters from all 390 in-sample applicants for the follow-up job, we proxy application effort by cover letter length. Appendix Figure A12 shows no treatment effects on cover letter length, suggesting that changes in application effort do not explain

the observed patterns in wage bids.

Second, since we observe only the wage bids submitted at the application stage, novices may strategically bid low to attract attention and renegotiate after receiving an offer. To assess this, we randomly offered the job to 34 in-sample novices and tracked offer acceptance, as described in Section 3.3. We find no evidence of such strategic bargaining. All but one freelancer accepted the offer at their proposed wage and completed the task; the remaining freelancer declined due to time constraints.

A final concern is that novices who undercut may exert less effort, leading to lower-quality output. We test this using worker performance on the follow-up task. Appendix Figure A13 Panel A shows that freelancers who bid lower wages performed, on average, better than those who did not undercut, although the difference is not statistically significant. Panel B reports task performance by treatment group and similarly shows no statistically significant differences. These patterns are consistent with novices using low bids to secure initial jobs while maintaining effort to build their reputation. Given the small sample size ($n = 33$), however, these findings should be interpreted as suggestive rather than causal.

6.3 Mechanisms

6.3.1 What Drives the Heterogeneity?

The heterogeneous treatment effects by novice ability are consistent with predictions from the standard reputation model in Tirole (1988), but raise a question: if most workers believe they are high performers at baseline, why does feedback induce high types to undercut when they presumably already know their quality? To answer this, we examine heterogeneity in the accuracy and confidence of novices' beliefs about their own performance. We measure accuracy as the absolute difference between perceived and actual baseline performance, and confidence as workers' self-reported confidence in the accuracy of their self-assessment, measured in the baseline and endline surveys.²¹ Appendix Table A7 reports the results.

Panel A, columns 1 to 4 show that the treatment effects for high-type novices are concentrated among those with more accurate priors in the feedback-only treatment relative to the control group. This pattern partly reflects a mechanical relationship: since most workers perceive themselves as high performers, true high types tend to have more accurate priors. Columns 5 and 6 split the sample by whether confidence in the accuracy of own performance assessment increases from baseline to endline. The effects are concentrated among workers whose confidence increases. This suggests that feedback primarily validates high types' prior beliefs rather than shifting them.

²¹In both baseline and endline surveys, novices were asked to evaluate their own performance on the baseline task and indicate how confident they are in their guess, on a scale of one to five.

An alternative explanation is that high-type novices respond more because they better understand the feedback. We test this using Figure 6, Panel A, which compares changes in workers’ beliefs about their own performance from baseline to endline. While high-performing novices’ beliefs move closer to the truth in the feedback group, low-performing novices exhibit larger updates from baseline. This indicates that low-type novices also process the feedback and adjust their beliefs about their own performance, which is inconsistent with differential comprehension as the primary mechanism.

6.3.2 Is There Complementarity Between Treatments?

Since the interventions target different sources of misbeliefs, a natural question is why the combined treatment exhibits limited complementarity. We use baseline data on workers’ beliefs about bidding strategies to address this. Appendix Table A8 reports the results. Columns 1 and 2 show that the treatment effects for high-ability novices in the feedback group are concentrated among those who initially hold more accurate beliefs that employers do not interpret low prices as signals of low quality. Columns 3 to 6 further show that these workers are more likely to believe that undercutting is effective for high types.

These patterns yield two insights. First, high-ability workers understand which strategies are effective but remain uncertain about their own ability. The feedback treatment therefore reinforces their self-assessment and induces them to adopt the undercutting strategy they already believe is effective. Second, because these workers hold correct priors about employer responses, the second treatment adds little, leading to limited complementarity.

In summary, the interventions address belief frictions through different channels. Correcting workers’ misperception about employer responses to low wage bids encourage those with inaccurate beliefs about the demand side to offer low wages to secure jobs. Resolving workers’ uncertainty about own abilities strengthens high-ability workers’ incentives to invest in reputation through low wage offers. This improves the allocation of talent by encouraging more efficient sorting across novice types.

7 Discussion

7.1 How Fast Can Novices Learn about Demand?

Our experimental results show that novice freelancers often avoid lowering their initial wage bids because they hold incorrect beliefs that employers interpret low prices as signals of low quality. This raises an important policy question: can novices eventually overcome these belief frictions through trial and error on the platform, or is external intervention needed to correct these misperceptions? To shed light on this question, we simulate an ideal

learning environment in which a fully rational novice experiments systematically with wage bids and updates her beliefs about demand according to Bayes rule. This exercise provides a conservative benchmark for how long and how costly the learning process can be in the absence of external intervention.²²

We model a novice freelancer conducting an A/B test by alternating between two bidding strategies when submitting job applications: bidding at the full budget or undercutting by 20 percent. Each application yields a binary outcome indicating whether the novice receives a callback. The novice starts with a 50/50 prior over two possible states of the world:

Model 1 (“Low wage bids $\neq$ low quality”): Employers do not penalize low bids. Undercutting by 20 percent increases callback rates from 3.41% to 5.26% for novices.

Model 2 (“Low wage bids = low quality”): Employers respond less to novices who submit lower bids. The callback rate at the full budget is 3.41%. We consider two degrees of misbelief about undercutting. Under moderate misbeliefs, novices believe the callback rate declines by 20 percent, from 3.41% to 2.73%. Under strong misbeliefs, they believe the callback rate declines by 50 percent, to 1.71%.

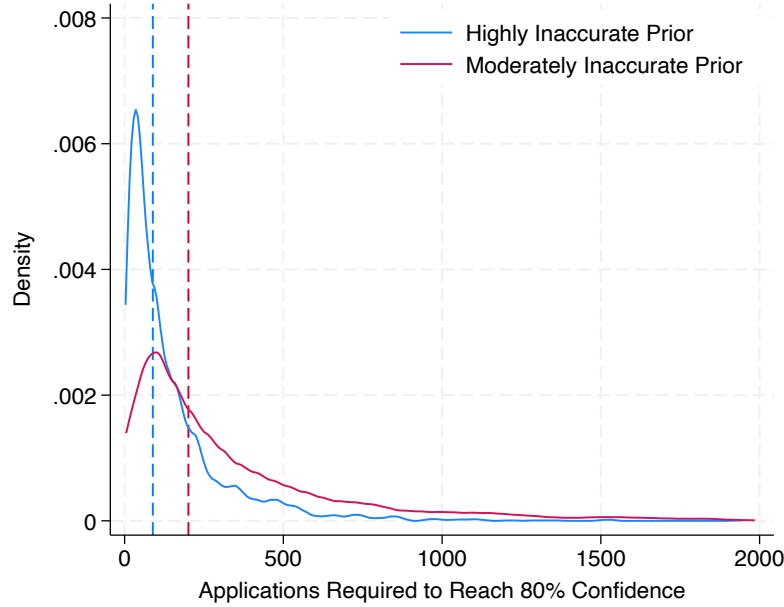
After each application, the novice updates beliefs about the two models using Bayes’ rule based on whether a callback is received. Learning is complete once the posterior probability assigned to the true model reaches 80%. We simulate 3,000 novice workers, each submitting up to 2,000 applications, and record the number of applications required to reach this threshold.

The simulation reveals that even under systematic experimentation and perfect Bayesian updating, learning is slow. Figure 8 plots the distribution of applications required to learn under the two types of inaccurate priors. Over 99% novices reach 80% certainty of the true model within 2,000 applications. The median worker with moderately inaccurate beliefs needs 201 applications to reach 80% confidence, while with strongly inaccurate beliefs needs 89 applications. Learning is faster under stronger misbeliefs because each successful application is more informative about the true model and leads to belief update. Nevertheless, learning remains slow overall because callbacks are rare and most applications provide little information. As a result, even perfectly rational workers cannot quickly infer how employers respond to wage undercutting.

The learning process translates into substantial real-world costs. Novice freelancers in our baseline survey submit on average 10 applications per month at an average cost of \$1.35 USD per application (excluding workers’ effort costs). A novice with moderately inaccurate

²²This simulation focuses on learning about employer responses to wage bids. Novices face a similar learning challenge in assessing their own abilities, as they need to secure jobs and receive employer feedback to update beliefs about their performance. Therefore, learning about one’s ability would impose similar (if not more) time and financial costs as learning about demand.

Figure 8: Distribution of Applications Required to Learn True Demand



Notes: The figure plots the simulated distribution of applications required for novice freelancers with inaccurate prior beliefs to learn the true demand model (model 1). Results are based on 3,000 simulated workers who alternate between high and low bids and update their beliefs using Bayes' rule. The dotted line indicates the median number of applications needed to reach 80% confidence in the true model.

beliefs would need 20 months of experimenting at the cost of \$271 USD, which is equivalent to 24% of the average annual income in low-income countries according to the World Bank.

These costs help rationalize why belief frictions may persist in practice and point to the importance of external interventions that reduce the cost of learning, rather than relying on workers to discover these patterns on their own. Even under ideal learning conditions where workers experiment systematically and update beliefs perfectly, the time and financial costs of experimentation are substantial. In reality, workers may experiment less frequently or update beliefs imperfectly, making learning even slower. As a result, many novices may never experiment enough to discover that lowering wage bids improves hiring chances, preventing them from adopting effective bidding strategies to overcome entry barriers in online labor markets. To facilitate worker learning and market entry, existing government initiatives, such as the Ajira Digital Program in Kenya and the Mastering the World of Online Freelancing projects in Jordan and Lebanon, incorporate mentorship and business strategy guidance (Datta et al., 2023). Similar components could be integrated into digital training programs to help workers form accurate expectations about job search in global markets.

7.2 Generalization to Other Settings

Next, we discuss how our findings translate to other labor market settings, especially in developing countries.

7.2.1 Online Labor Markets

Our analysis focuses on data entry services on one of the largest online freelancing platforms, so a natural question is how much the entry barriers and frictions we document extend to other service categories and platforms. As discussed in Section 3.2, similar entry barriers arise broadly: inexperienced workers struggle to get hired because employers are uncertain about their ability.

One potential difference is that workers in higher-skill categories, such as graphic design or programming, may have access to alternative signals of ability, such as portfolios or certifications, rather than relying on lower initial wages. However, these signals may be less informative than they appear. Using profile data from 3,461 graphic designers on the same platform, we find that 96% list a portfolio and 40% report certifications. When such signals are nearly universal, they provide limited differentiation across workers. This pattern is consistent with [Kässi and Lehdonvirta \(2022\)](#), who find that micro-credentials have little effect on job outcomes for new entrants.

There is also evidence that introductory pricing is used in higher-skill categories. [Cui \(2025\)](#) shows that freelancers specializing in PHP adopt low initial prices to build reputation on Freelancer.com. At the same time, these investments remain below the social optimum, leading to inefficient hiring of novice workers. This suggests that the underuse of introductory pricing is not unique to low-skill services.

A remaining question is whether the belief frictions we document are present in these settings. There is limited direct evidence, as our study is the first to elicit freelancers' beliefs about their own ability and employer responses. That said, similar frictions have been documented in offline labor markets, and there are reasons to expect them to persist online. In particular, novices from developing countries may have limited information about their standing in the global talent pool or about preferences of foreign employers. Platform design may further slow learning: on many platforms, workers cannot observe competing bids, making it difficult to infer demand. In contrast, platforms that reveal bid ranges may facilitate faster learning and stronger price competition.

7.2.2 Offline Labor Markets in Developing Countries

While this paper studies online marketplaces, our results are relevant in offline labor markets. These markets share three features. First, both settings resemble spot markets, characterized

by high unemployment, high search frictions, and short job spells (Breza and Kaur, 2025). Wages in these markets exhibit limited dispersion relative to underlying productivity (Breza, Kaur and Shamdasani, 2018; Breza, Kaur and Krishnaswamy, 2019; Cefala et al., 2024; K, 2025). Second, inexperienced workers face substantial barriers to entry. Because educational credentials often provide weak signals of ability, novices struggle to credibly convey their productivity to employers (Bertrand et al., 2024). As a result, employers rely heavily on observable work experience as a proxy for quality.²³ Third, workers often have limited information about both their own abilities and employer preferences (Carranza et al., 2022; Kiss et al., 2023; Caria et al., 2024). Together, these features suggest that supply-side belief frictions may contribute to compressed wages and inefficient allocation, particularly for high-ability novices.

At the same time, three features of offline labor markets may limit the direct applicability of our results. First, the scope for reputation building may be more limited: job spells are typically short, and past performance is often difficult to verify across employers (Donovan, Lu and Schoellman, 2023). This weakens workers’ incentives to accept lower initial wages as an investment to reveal their true ability. However, when workers remain with the same employer, returns to experience can be substantial (Breza and Kaur, 2025) because worker performance information becomes available over time. This mirrors the reputation-building process on online platforms, where performance history is available to all potential employers. In both settings, high-ability workers have incentives to invest in reputation, improving match quality. Second, subsistence constraints may restrict workers’ ability to undercut wages. Nevertheless, recent evidence shows that wage undercutting does occur even among low-wage manual laborers in rural settings (Breza, Kaur and Krishnaswamy, 2019; K, 2025), indicating that workers engage in strategic wage-setting despite financial constraints. Third, wage undercutting may be discouraged by social norms in rural labor markets, even in the absence of belief frictions (Breza, Kaur and Krishnaswamy, 2019). As a result, the mechanisms we document are likely to be most relevant in urban labor markets and sectors with higher returns to experience, such as services and manufacturing, where both reputation and strategic pricing play a larger role.

8 Conclusion

This paper studies why novice workers from LMICs under-adopt introductory pricing to overcome entry barriers in online labor markets. In the demand-side experiment, we show that employers respond favorably to low wage bids from novices and do not interpret them as signals of low ability. Back-of-the-envelope calculations further indicate that undercutting

²³For example, evidence from Burundi shows that employers prefer candidates with prior experience in low-skill jobs over otherwise identical inexperienced applicants (Ndayikeza, 2025).

is a profitable entry strategy.

Despite these returns, few novices adopt it. Survey evidence points to two frictions: incorrect beliefs about how employers interpret low wages and limited information about their own ability. In a supply-side experiment, we show that providing workers with information about employer responses and their relative performance increases novices' likelihood of proposing low initial wages. The effect is driven by high-ability novices after receiving performance feedback, consistent with standard reputation models. This separation in wage bids improves talent allocation by increasing high-ability novices' chances of being hired and producing high-quality output.

We then examine whether novices can correct their misbeliefs through on-platform search and application. Simulations show that even under Bayesian updating, learning is slow and costly, as most applications receive no response and thus offer little scope for learning. This highlights the value of external information provision, such as mentorship or guidance on bidding strategies, in helping novices overcome entry barriers. While we do not observe how employers use wage bids or other profile signals to infer worker quality, better information on employer decision-making would help high-ability novices optimize their application strategies and improve entry into the market.

Nevertheless, the implications of these findings can be extended beyond online labor markets to broader questions of information provision and market efficiency. Our results suggest that policymakers can improve talent discovery and worker outcomes by providing skill assessments for new entrants and sharing information about employer decision-making processes. More broadly, the findings indicate that policies addressing information frictions on the supply side, rather than the demand side, may be more effective for improving job matching in markets characterized by high entry barriers and low-information environments.

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Appendix A Figures and Tables

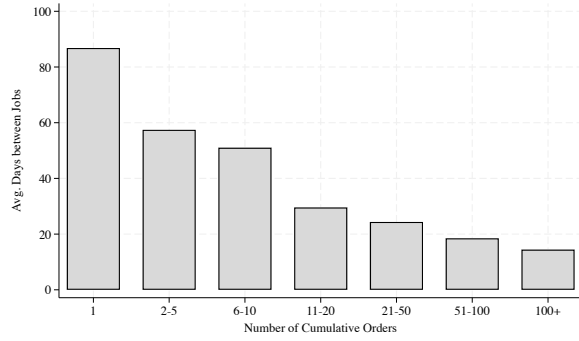
Appendix Figure A1: Freelancer Profiles for Audit Study

 Pakistan Data Cleaning & Database Management Expert   Message Hire	0 completed jobs 0 total hours \$0 earned	12 skills on their profile Data Analysis Online Research Data Scraping Data Extraction Lead Generation Microsoft Word Data Entry Data Mining +4	\$40.00 Cover letter - Hi, I'm writing in response to your job post for Data extraction for online reports. I have experience in this area and would be glad to assist with compiling data with accuracy. I understand the importance of accuracy and timely delivery, an... View more
 Pakistan E-commerce Data Entry & Product Listing Specialist   Message Hire	100% Job Success 4 completed jobs 0 total hours \$60 earned	12 skills on their profile Microsoft Word Microsoft Excel Lead Generation Customer Service Data Mining Data Extraction +6	\$40.00 Cover letter - Hi there, I saw your job titled Data extraction for online reports and thought I'd reach out. I've worked a lot with data extraction projects and would be happy to help you with this job! I have completed multiple data entry and research... View more
 Pakistan Accurate & Fast Typist Data Entry & Transcription   Message Rehire	100% Job Success 4 completed jobs 0 total hours \$50 earned	12 skills on their profile Data Analysis Online Research Data Scraping Customer Service Data Extraction Lead Generation Microsoft Word +5	\$32.00 Cover letter - Dear Sir/Madam, I saw your post for Data extraction for online reports. I believe I am the perfect candidate to help you with data scraping and collection. I have worked on a range of data entry and research projects on Upwork with... View more
 Pakistan Data Entry & Admin Support   Message Hire	0 completed jobs 0 total hours \$0 earned	12 skills on their profile Lead Generation Data Analysis Online Research List Building Microsoft Excel Microsoft Word Virtual Assistance +5	\$32.00 Cover letter - Dear Sir/Madam, I came across your job for Data extraction for online reports and would love to assist you with extracting data from financial reports and entering them into your Excel template. With a strong background in data entry an... View more

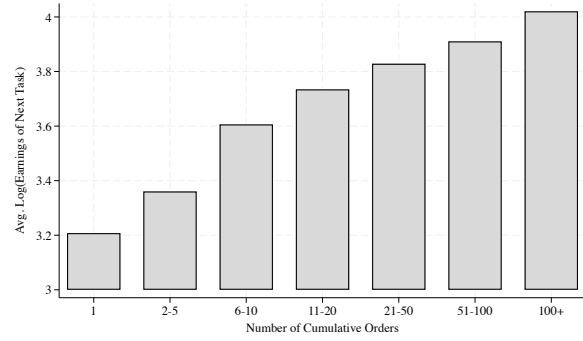
Notes: This is a screenshot taken from the employer's dashboard on the online freelancing platform. The original job post has a fixed budget of 40 USD. The applications are sent by four freelancer profiles in the audit study. The top and the bottom applications are sent by inexperienced freelancers and the middle ones by experienced freelancers.

Appendix Figure A2: Hiring Prospects and Completed Jobs

Panel A. Gap between Jobs (Days)

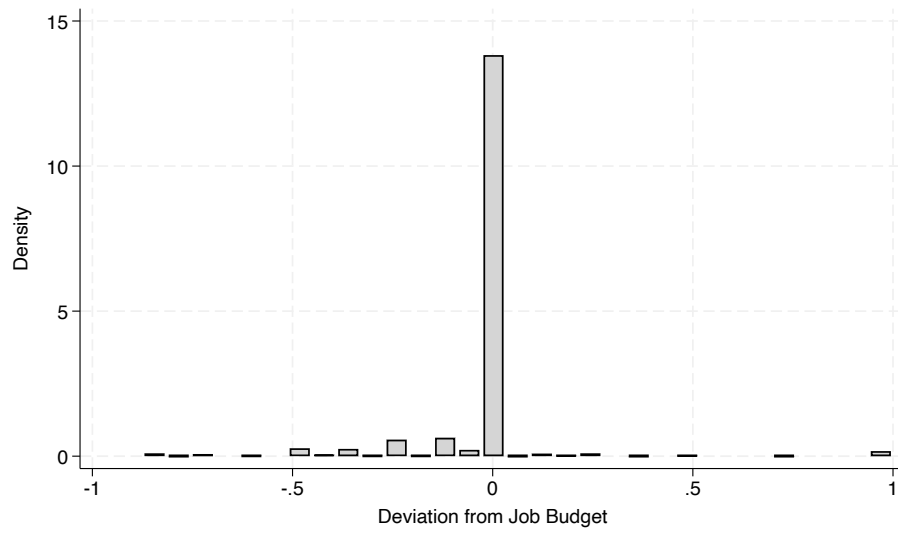


Panel B. Log Earnings of Future Jobs



Notes: The figures plot the average number of days between jobs (Panel A) and the log of earnings from the subsequent job (Panel B) against the number of completed jobs. The job-level data was collected for all active Pakistani workers specialized in data entry on the studied platform by June 2024.

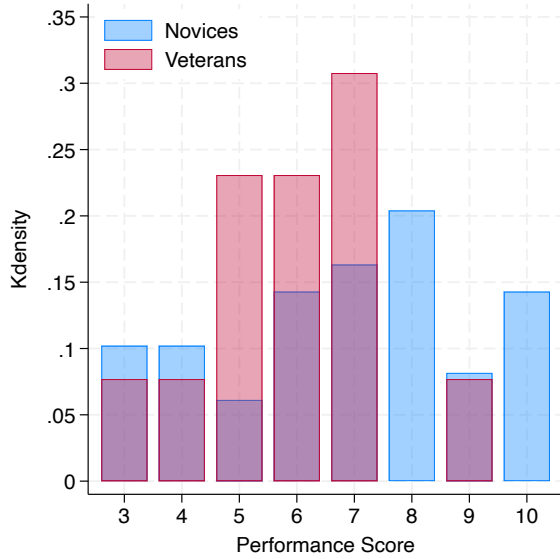
Appendix Figure A3: Distribution of Deviation in Proposed Wages



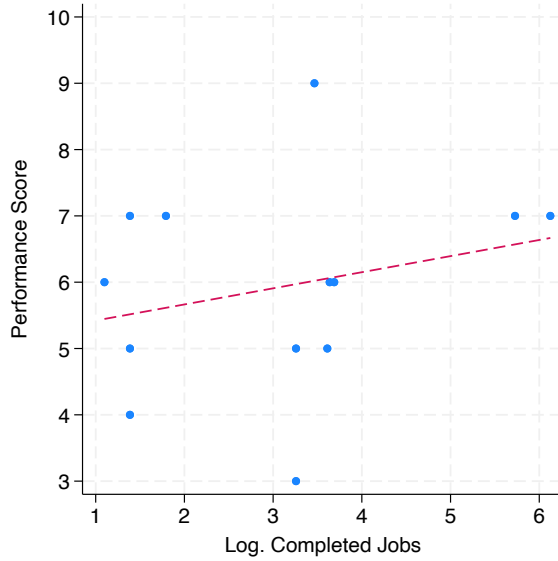
Notes: This figure shows the distribution of deviation in proposed wages from the job budget, measured in the application data.

Appendix Figure A4: Follow-up Task Performance

Panel A. Novices versus Veterans

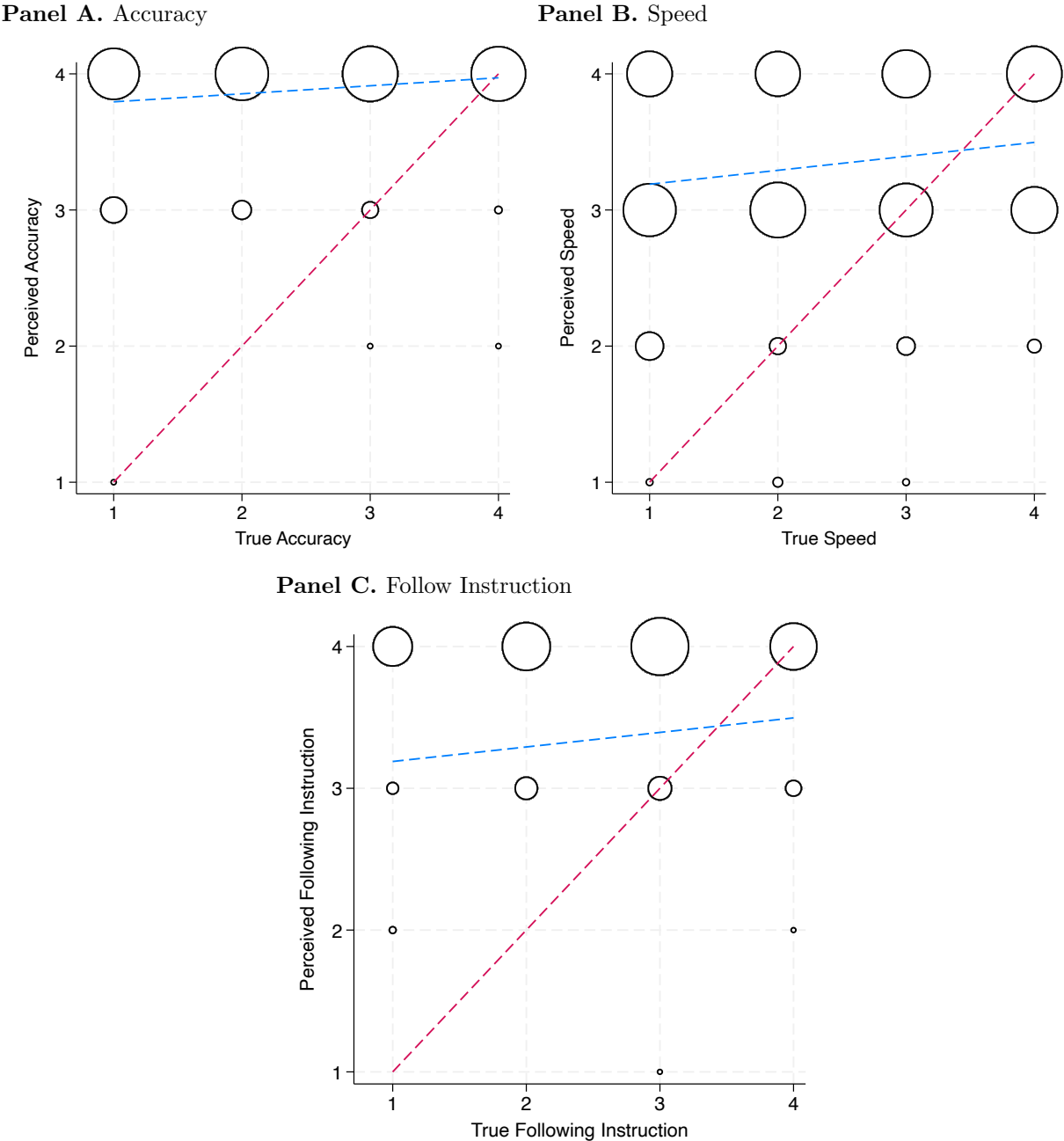


Panel B. Veteran Performance by Experience



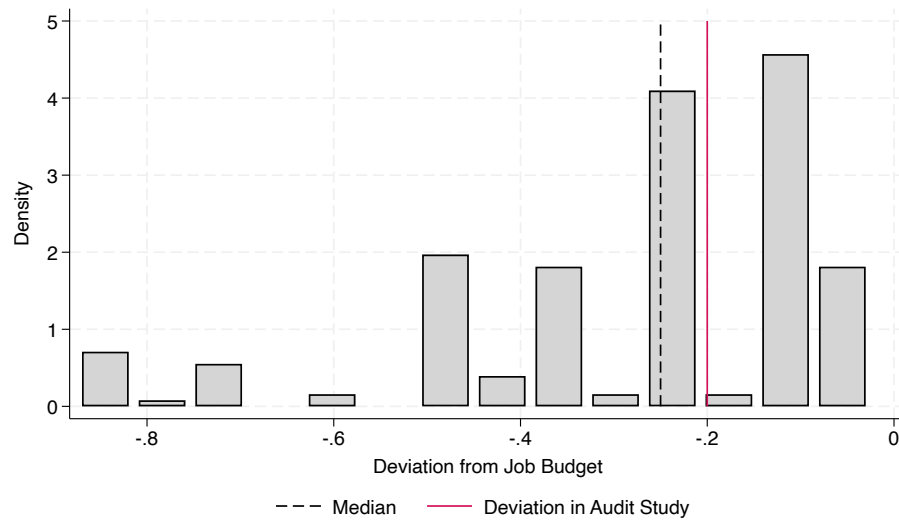
Notes: This figure describes follow-up task performance. Panel A plots the distribution of follow-up performance scores for novices ($n = 49$) and veterans ($n = 13$). Performance score ranges from 3 to 10, computed as the sum of quartile ranking in accuracy and speed, and a binary variable in whether workers followed task instructions perfectly. Panel B plots the relationship between log completed jobs and performance scores among veteran workers.

Appendix Figure A5: Actual versus Perceived Performance by Each Dimension



Notes: This figure shows the actual and perceived scores for accuracy, speed, and following instructions, with the true scores on the x-axis and the perceived scores on the y-axis. The red dotted line is the 45-degree line and the blue line shows the best linear fit. The marker size is relative to the number of observations.

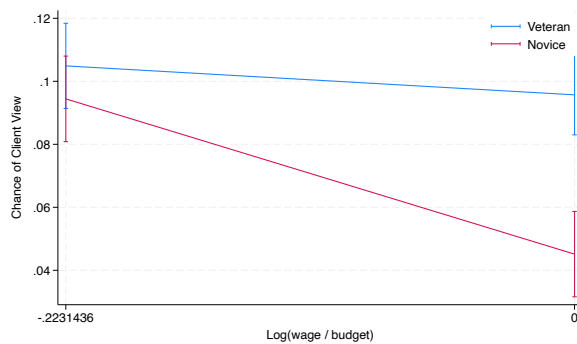
Appendix Figure A6: Distribution of Wage Bids Below Job Budget



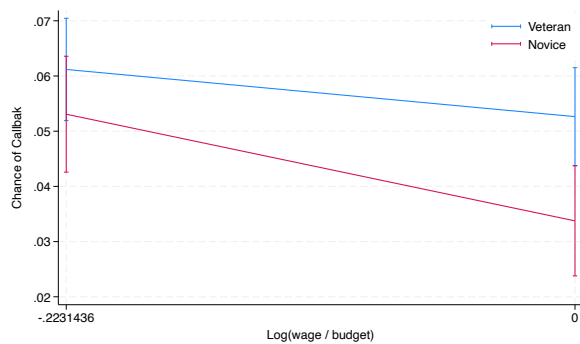
Notes: This figure shows the distribution of wage bids below the job budget, measured in the application data. The dotted black line shows the median value of the deviation and the solid red line represents wage undercutting by 20%, which is what we proposed in the audit study.

Appendix Figure A7: Demand Response

Panel A. Employer View

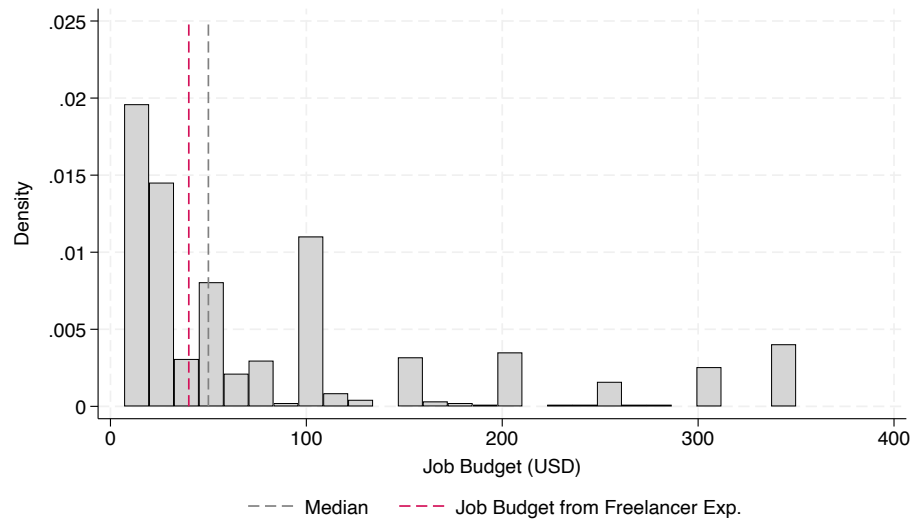


Panel B. Callback



Notes: This figure plots mean employer views (Panel A) and callbacks (Panel B) by wage bid and worker experience. The horizontal axis shows $\log(\text{wage}/\text{budget})$, where 0 corresponds to bidding at the posted budget and 0.223 corresponds to bidding at 80% of the budget. Vertical bars denote 95% confidence intervals.

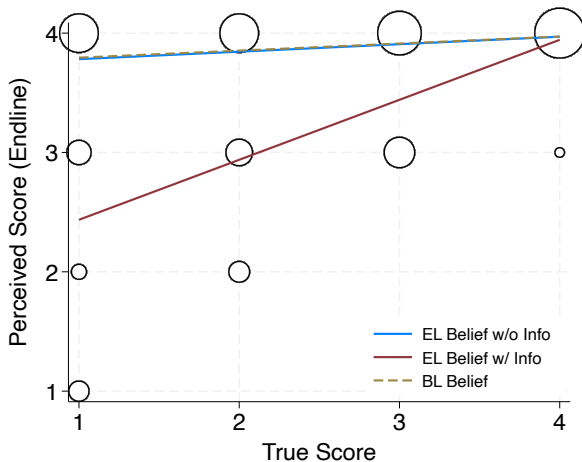
Appendix Figure A8: Distribution of Budget for Data Entry Jobs



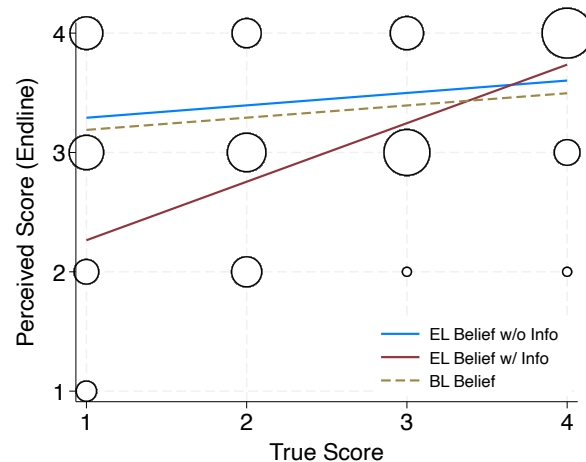
Notes: This figure shows the distribution of job budget from 703 data entry jobs in the demand experiment. The red dotted line shows where the job budget for the referred job falls in the freelancer experiment.

Appendix Figure A9: Treatment Effects on Worker Beliefs by Performance Dimension

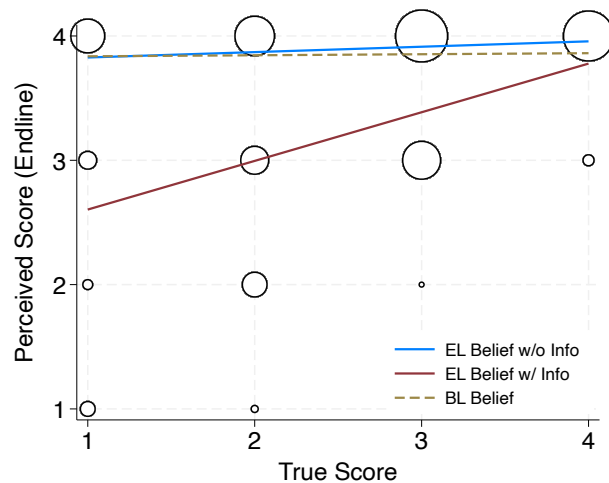
Panel A. Accuracy



Panel B. Speed

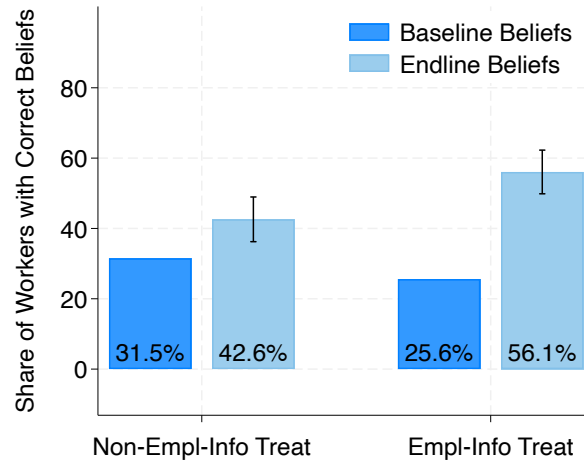


Panel C. Follow Instruction



Notes: This figure plots workers' actual score for each performance dimension (accuracy, speed, and following instruction) against their perceived score measured at the endline. The blue line represents the best linear fit for workers without performance feedback and the red line for workers with performance feedback. The dotted line shows the best linear fit of baseline perceived performance against true performance.

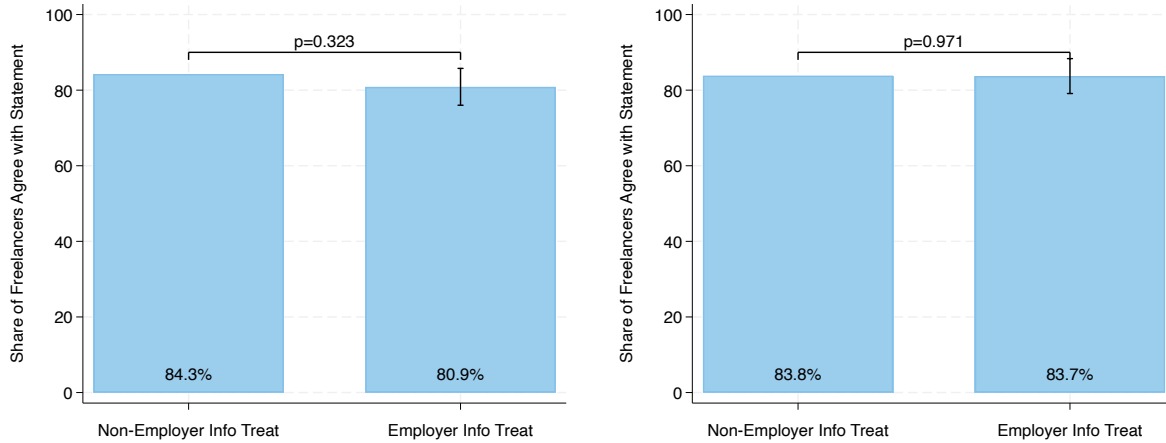
Appendix Figure A10: Baseline and Endline Beliefs by Treatment



Notes: This figure plots the share of workers who “disagree” or “strongly disagree” with the statement in the baseline survey: “If a new freelancer offers a lower price, the client may think that their work is low quality.”; and those who “agree” or “strongly agree” with the statement in the endline survey: “The employer (of the referred job) will not judge my quality based on my wage bids.” We report the 95% confidence intervals and p-values from the Wald test.

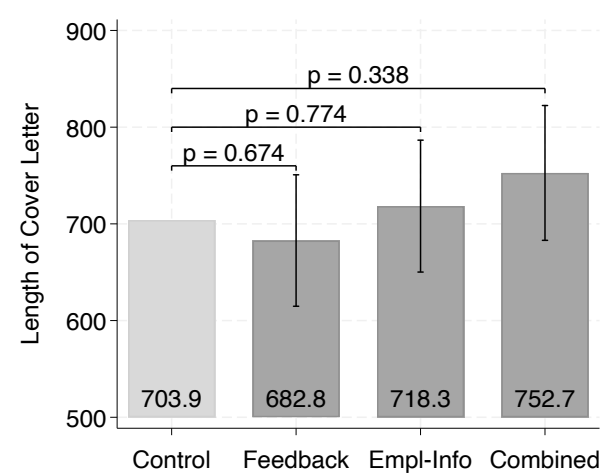
Appendix Figure A11: Treatment Effect on Worker Perceptions of Referred Job

Panel A. This job opportunity seems trustworthy. **Panel B.** Competition may be strong.



Notes: This figure reports the treatment effects on worker beliefs about the trustworthiness and competition for the referred job. We report the 95% confidence intervals in both panels and p-values from the Wald test.

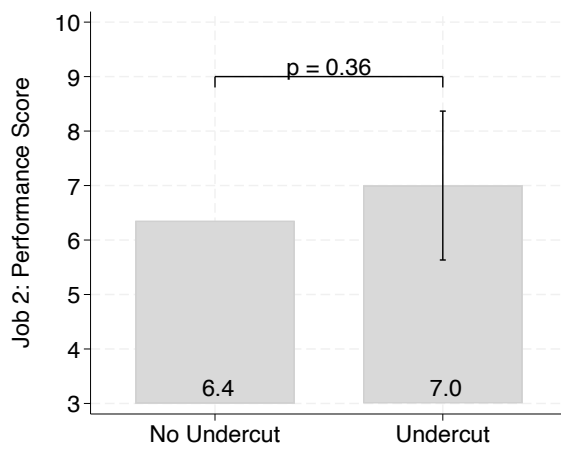
Appendix Figure A12: Treatment Effect on Cover Letter Length



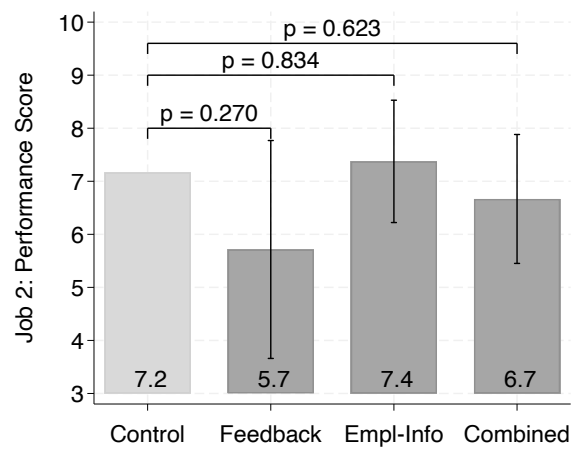
Notes: This figure reports the effects of treatments on the cover letter length by in-sample novices ($N = 390$) who applied to the follow-up task. We report the 95% confidence intervals and p-values from the Wald test in both panels.

Appendix Figure A13: Treatment Effect on Follow-up Task Performance

Panel A. Performance by Whether Undercut



Panel B. Performance by Treatment



Notes: This figure reports the effects of undercutting and treatments on performance (score between 3 and 10) by in-sample novices who completed the follow-up task. We report the 95% confidence intervals and p-values from the Wald test in both panels.

Appendix Table A1: Job and Employer Characteristics

	Median	Mean	Min	Max	N
<i>Job Characteristics</i>					
Job Budget (USD)	50	112	7	8,000	743
Any Freelancer Hired	1.00	0.54	0.00	1.00	743
No. Freelancer Hired	1.00	0.73	0.00	14.00	743
<i>Employer Characteristics</i>					
Employer Tenure (Month)	24	40	0	281	720
Employer from HICs	1.00	0.85	0.00	1.00	720
No. Jobs Posted	21	139	1	3,468	720
Employer Rating (Out of 5)	4.99	4.86	1.75	5.00	565

Notes: This table reports characteristics of jobs and employers in the analysis sample of the employer experiment. The first panel shows job characteristics and hiring outcomes indicated on the platform, at least two weeks after the initial job post. The second panel shows employer characteristics. *Employer Tenure* is measured by the gap between when the employer first joined the platform and when the job was posted. *Employer from HICs* indicates whether the employer is located in a high-income country, using World Bank country classifications by income level for 2024-2025. *No. Jobs Posted* refers to the total number of jobs the employer has posted since joining the platform. *Employer Rating* is a measure of reputation (from one to five) generated by the platform based on rating by freelancers previously hired by the employer. New employers without any hiring history do not have a rating.

Appendix Table A2: Type of Callback Messages

Type	Count	Share (%)
Request for additional info	32	22.53
Task instructions	30	21.13
Request for work samples	29	20.42
Confirm interest	21	14.79
Interview request	16	11.27
Other	13	9.15
Job offer	1	0.70
N	142	

Notes: This table categorizes callback messages received during the employer experiment. Shares are reported as percentages of the total number of callbacks ($N = 142$). “Request for additional information” refers to employers asking applicants to provide details such as availability or relevant experience. “Task instructions” refers to employers sharing details about the assignment. “Request for work samples” includes requests for past work samples or for applicants to complete a sample task. “Confirm interest” refers to employers checking whether the freelancer remains interested in the position. “Interview request” refers to invitations to interview. “Other” includes messages that do not fall into the preceding categories, such as a greeting. “Job offer” refers to direct offers of employment for the posted position.

Appendix Table A3: Treatment Effects on Employer Responses

	Client Viewed				Contacted			
	Analysis Sample		Full Sample		Analysis Sample		Full Sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Novice (β_1)	-0.050 (0.011***)	-0.043 (0.013***)	-0.061 (0.015***)	-0.041 (0.012***)	-0.018 (0.008**)	-0.025 (0.009***)	-0.021 (0.010**)	-0.025 (0.009***)
Undercut (β_2)	0.010 (0.010)	0.009 (0.010)	0.009 (0.009)	0.010 (0.010)	0.009 (0.007)	0.008 (0.007)	0.008 (0.006)	0.008 (0.006)
Novice $\times$ Undercut (β_3)	0.038 (0.015**)	0.039 (0.015**)	0.051 (0.017***)	0.035 (0.015**)	0.010 (0.011)	0.010 (0.011)	0.014 (0.012)	0.010 (0.010)
Constant	0.095	0.082	0.114	0.080	0.053	0.050	0.057	0.049
Test: $\beta_2 + \beta_3 = 0$	0.000	0.000	0.000	0.000	0.024	0.026	0.026	0.019
Linear Prob.	$\times$	$\times$		$\times$	$\times$	$\times$		$\times$
Logit			$\times$				$\times$	
Job FE	$\times$	$\times$		$\times$	$\times$	$\times$		$\times$
Profile FE		$\times$		$\times$		$\times$		$\times$
No. Applications	2812	2812	2812	2972	2812	2812	2812	2972

Notes: This table reports treatment effects on employer views and callbacks by worker experience and wage undercutting. Columns (1) and (5) present the baseline specification corresponding to Figure 3. Columns (2) and (6) include freelancer profile fixed effects. Columns (3) and (7) report logit estimates. Columns (4) and (8) extend the original analysis to the full sample with profile and job fixed effects. The omitted group is veteran applicants without wage undercutting. We report p-values from the statistical test for $\beta_2 + \beta_3 = 0$ from all specifications. Standard errors are clustered at the job level. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Appendix Table A4: Heterogeneous Treatment Effects on Employer Responses

	Job Budget		Employer Tenure		HIC Employer		No. Jobs Posted		Employer Rating	
	< Median (1)	≥ Median (2)	< Median (3)	≥ Median (4)	= 0 (5)	= 1 (6)	< Median (7)	≥ Median (8)	< Median (9)	≥ Median (10)
<i>Panel A: Employer View</i>										
Novice (β_1)	-0.057 (0.016***)	-0.043 (0.016***)	-0.074 (0.018***)	-0.023 (0.013*)	-0.029 (0.036)	-0.052 (0.012***)	-0.083 (0.018***)	-0.014 (0.014)	-0.030 (0.018)	-0.073 (0.017***)
Undercut (β_2)	0.006 (0.014)	0.013 (0.016)	0.003 (0.016)	0.014 (0.014)	0.000 (0.024)	0.010 (0.012)	0.006 (0.016)	0.011 (0.013)	0.015 (0.017)	-0.015 (0.015)
Novice $\times$ Undercut (β_3)	0.039 (0.020*)	0.038 (0.023)	0.057 (0.024**)	0.023 (0.020)	0.049 (0.043)	0.038 (0.017**)	0.074 (0.024***)	0.006 (0.020)	0.023 (0.026)	0.055 (0.024**)
Omitted Mean	0.099	0.092	0.116	0.075	0.137	0.088	0.122	0.069	0.098	0.099
No. Applications	1328	1484	1344	1392	408	2328	1344	1392	1064	1092
<i>Panel B: Employer Contact</i>										
Novice (β_1)	-0.021 (0.012*)	-0.016 (0.011)	-0.030 (0.013**)	-0.003 (0.010)	0.000 (0.024)	-0.019 (0.009**)	-0.030 (0.012**)	-0.003 (0.011)	-0.011 (0.016)	-0.029 (0.011**)
Undercut (β_2)	0.018 (0.011)	0.000 (0.008)	0.006 (0.011)	0.014 (0.008*)	0.000 (0.014)	0.012 (0.007)	0.009 (0.009)	0.011 (0.010)	0.011 (0.012)	0.015 (0.010)
Novice $\times$ Undercut (β_3)	-0.003 (0.014)	0.022 (0.016)	0.018 (0.017)	-0.003 (0.014)	0.029 (0.030)	0.003 (0.012)	0.021 (0.015)	-0.006 (0.016)	-0.004 (0.022)	0.004 (0.012)
Omitted Mean	0.057	0.049	0.065	0.037	0.088	0.045	0.057	0.046	0.071	0.044
No. Applications	1328	1484	1344	1392	408	2328	1344	1392	1064	1092

Notes: This table examines the heterogeneity in employer's response to wage undercutting by novice and veteran freelancers. For *Job Budget*, *Employer Tenure*, *No. Jobs Posted*, and *Employer Rating*, we divide the analysis sample by the median and for *HIC Employer*, the analysis sample is split by whether the employer is located in a high-income country, using World Bank country classifications by income level for 2024-2025. Panel A reports the heterogeneous treatment effects on whether employers viewed the application and Panel B on whether employers reached out to the applicant. All regressions control for job fixed effects. The omitted group is applications by veteran workers without wage undercutting. Standard errors are clustered at the job level. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Appendix Table A5: Data Sources for Back-of-the-Envelope Calculation

	Parameter Values		Source
	Bid 80%	Bid Full	
Contact Probability (p)	5.26%	3.41%	Demand Experiment
Initial Job Earnings (π_1)	50	40	Demand Experiment
Application Fee (c_a)	1.35	1.35	Demand Experiment
Earning Growth by Completed Jobs	-	-	Job History Data
Application Sent per Month	10	10	Freelancer Survey
Job Effort Cost ($c_e = w * h$)	12.5	12.5	Freelancer Survey
Hourly reservation wage (w)	5	5	Freelancer Survey
Hours taken per task (h)	2.5	2.5	Freelancer Survey

Notes: This table reports the key parameters used in the back-of-the-envelope calculation of the returns to wage undercutting, along with their data sources. The initial job earnings reflect the original job budgets posted by employers before any wage discounting. These values differ across bidding strategies because, in the demand experiment, jobs that responded to lower bids had a higher median budget than those that responded to full-price bids. Earnings growth is computed as a function of the number of completed jobs using the job history data, so no single value is reported here. Application behavior and effort costs are drawn from the baseline freelancer survey. The job effort cost equals the product of novices' self-reported reservation wage and the hours required to complete their first platform job.

Appendix Table A6: Balance Test for Freelancer Experiment

	All	Control	Info		Signal		Info+Signal		p-val
N	481	118	117		126		120		
	Mean (SD)	Mean (SD)	Mean (SD)	β_1 (p-val)	Mean (SD)	β_2 (p-val)	Mean (SD)	β_3 (p-val)	
<i>Panel A: Personal Characteristics</i>									
Female	0.52 (0.50)	0.56 (0.50)	0.55 (0.50)	-0.01 (0.86)	0.47 (0.50)	-0.09 (0.16)	0.50 (0.50)	-0.06 (0.36)	0.45
Age	28.76 (5.98)	29.16 (6.67)	28.55 (5.87)	-0.62 (0.45)	28.96 (5.65)	-0.19 (0.81)	28.35 (5.74)	-0.81 (0.32)	0.72
Bachelor Degree or Above	0.83 (0.38)	0.79 (0.41)	0.85 (0.36)	0.06 (0.24)	0.83 (0.38)	0.04 (0.46)	0.85 (0.36)	0.06 (0.21)	0.59
Employed Offline	0.52 (0.50)	0.52 (0.50)	0.58 (0.50)	0.06 (0.32)	0.50 (0.50)	-0.02 (0.80)	0.48 (0.50)	-0.03 (0.61)	0.44
Full-Time Freelancer	0.36 (0.48)	0.35 (0.48)	0.35 (0.48)	0.00 (0.95)	0.33 (0.47)	-0.02 (0.71)	0.40 (0.49)	0.05 (0.40)	0.67
Platform Tenure (Month)	19.65 (25.70)	22.25 (34.46)	17.50 (19.90)	-4.75 (0.20)	19.00 (24.62)	-3.28 (0.39)	19.87 (21.51)	-2.36 (0.53)	0.59
Present on Other Platforms	0.44 (0.50)	0.47 (0.50)	0.40 (0.49)	-0.07 (0.24)	0.48 (0.50)	0.00 (0.98)	0.42 (0.50)	-0.06 (0.38)	0.52
Has Earned from Other Platforms	0.18 (0.39)	0.20 (0.40)	0.20 (0.40)	-0.01 (0.89)	0.19 (0.39)	-0.01 (0.80)	0.14 (0.35)	-0.06 (0.22)	0.56
High Type	0.50 (0.50)	0.47 (0.50)	0.53 (0.50)	0.06 (0.40)	0.49 (0.50)	0.02 (0.79)	0.49 (0.50)	0.02 (0.79)	0.86
<i>Panel B: Profile Characteristics</i>									
Number of Completed Jobs	0.13 (0.35)	0.13 (0.33)	0.14 (0.35)	0.01 (0.80)	0.09 (0.28)	-0.04 (0.32)	0.18 (0.41)	0.06 (0.24)	0.18
Hourly Rate (USD)	7.91 (5.06)	7.71 (4.59)	7.89 (5.15)	0.19 (0.77)	7.93 (4.99)	0.22 (0.72)	8.10 (5.53)	0.39 (0.56)	0.95
Low Income Country	0.09 (0.28)	0.09 (0.29)	0.09 (0.29)	-0.00 (0.99)	0.10 (0.29)	0.00 (0.97)	0.07 (0.25)	-0.03 (0.44)	0.80
Lower-Middle Income Country	0.78 (0.42)	0.74 (0.44)	0.75 (0.43)	0.02 (0.79)	0.78 (0.42)	0.04 (0.45)	0.84 (0.37)	0.10 (0.04**)	0.17
Upper-Middle Income Country	0.13 (0.34)	0.16 (0.37)	0.15 (0.36)	-0.01 (0.89)	0.13 (0.33)	-0.03 (0.45)	0.09 (0.29)	-0.07 (0.11)	0.32
Fluent in English	0.97 (0.18)	0.95 (0.22)	0.95 (0.22)	0.00 (0.99)	0.99 (0.09)	0.04 (0.05*)	0.97 (0.16)	0.03 (0.30)	0.07*
Portfolio	0.65 (0.48)	0.61 (0.49)	0.69 (0.46)	0.08 (0.20)	0.70 (0.46)	0.09 (0.16)	0.61 (0.49)	-0.00 (0.97)	0.28
Certificate	0.43 (0.49)	0.41 (0.49)	0.41 (0.49)	-0.00 (0.96)	0.50 (0.50)	0.08 (0.20)	0.37 (0.49)	-0.04 (0.52)	0.27
Verified Certificate	0.08 (0.28)	0.04 (0.20)	0.06 (0.24)	0.02 (0.55)	0.14 (0.35)	0.09 (0.01**)	0.09 (0.29)	0.05 (0.13)	0.06*
Number of Skills	11.12 (3.72)	11.48 (3.65)	10.74 (3.74)	-0.73 (0.13)	11.28 (3.66)	-0.20 (0.66)	10.97 (3.83)	-0.51 (0.29)	0.43
Employment History	0.95 (0.21)	0.97 (0.18)	0.97 (0.16)	0.01 (0.69)	0.95 (0.22)	-0.01 (0.58)	0.92 (0.28)	-0.05 (0.11)	0.23

Notes: This table summarizes personal (panel A) and profile (panel B) characteristics of all participants in the freelancer experiment. Personal characteristics are measured from the baseline survey and the data entry task. Profile characteristics are public information collected from freelancer profiles on the platform. We show means and standard deviations within treatment arms as well as coefficients and p -values on the treatment indicators from the regression: $Y_i = \alpha + \beta_1 \text{Info}_i + \beta_2 \text{Signal}_i + \beta_3 (\text{Info} + \text{Signal})_i + \eta_r + \varepsilon_i$. The last column reports a joint test of mean values across treatment groups. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Appendix Table A7: Heterogeneity by Beliefs of Ability

Dependent Var:	Abs(Actual-Perceive Performance Score)				Confidence in Perceived Score	
Wage Undercutting	1st Quartile (1)	2nd Quartile (2)	3rd Quartile (3)	4th Quartile (4)	Decreased (5)	Increased (6)
<i>Panel A: Feedback Only versus Control</i>						
Treated	-0.500 (0.368)	-0.000 (0.000)	-0.004 (0.092)	0.016 (0.064)	-0.012 (0.062)	-0.056 (0.057)
High-Type	-0.444 (0.373)	0.050 (0.051)	-0.067 (0.067)	0.000 (0.000)	-0.010 (0.063)	-0.056 (0.057)
Treated $\times$ High-Type	0.686 (0.382*)	0.108 (0.101)	0.226 (0.172)	0.000 (0.000)	0.158 (0.097)	0.286 (0.135**)
Omitted Mean	0.500	0.000	0.067	0.037	0.069	0.056
No. Obs	51	45	46	46	142	46
<i>Panel B: Employer Info Only versus Control</i>						
Treated	0.167 (0.100)	0.250 (0.159)	0.161 (0.115)	0.180 (0.095*)	0.175 (0.081**)	0.069 (0.135)
High-Type	-0.444 (0.370)	0.050 (0.051)	-0.067 (0.067)	0.000 (0.000)	-0.010 (0.063)	-0.056 (0.056)
Treated $\times$ High-Type	0.000 (0.000)	-0.133 (0.185)	-0.161 (0.115)	0.000 (0.000)	-0.048 (0.109)	0.112 (0.181)
Omitted Mean	0.500	-0.000	0.067	0.037	0.069	0.056
No. Obs	47	55	46	50	151	47

Notes: This table examines the heterogeneity in treatment effects by workers' accuracy of own ability assessment and confidence in accuracy of their guess. Panel A reports the heterogeneous treatment effects for feedback-only and control groups and Panel B on employer info-only and control groups. All regressions control for round fixed effects. The omitted group is low-type workers in the control group. Standard errors are clustered at the job level. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Appendix Table A8: Heterogeneity by Beliefs of Employer Response

Dependent Var: Wage Undercutting	Employer Think Low Price=Low Quality		High-Type Should Undercut		Undercut Works Better for High-Type	
	Disagree (1)	Agree (2)	Disagree (3)	Agree (4)	Disagree (5)	Agree (6)
<i>Panel A: Feedback Only versus Control</i>						
Treated	-0.040 (0.081)	0.131 (0.105)	0.062 (0.073)	-0.069 (0.094)	0.007 (0.076)	-0.049 (0.111)
High-Type	0.051 (0.088)	-0.091 (0.080)	0.050 (0.068)	-0.069 (0.094)	0.018 (0.081)	-0.071 (0.100)
Treated $\times$ High-Type	0.262 (0.116**)	-0.051 (0.131)	-0.041 (0.096)	0.373 (0.131***)	0.102 (0.108)	0.309 (0.148**)
Omitted Mean	0.040	0.091	0.000	0.111	0.034	0.111
No. Obs	110	78	84	104	109	79
<i>Panel B: Employer Info Only versus Control</i>						
Treated	0.240 (0.099**)	0.088 (0.094)	0.217 (0.088**)	0.122 (0.098)	0.201 (0.086**)	0.099 (0.111)
High-Type	0.051 (0.102)	-0.091 (0.099)	0.050 (0.091)	-0.069 (0.104)	0.018 (0.101)	-0.071 (0.105)
Treated $\times$ High-Type	-0.170 (0.138)	0.130 (0.136)	-0.172 (0.126)	0.079 (0.140)	-0.078 (0.129)	0.075 (0.159)
Omitted Mean	0.040	0.091	0.000	0.111	0.034	0.111
No. Obs	103	95	84	114	122	76

Notes: This table examines the heterogeneity in treatment effects by workers' baseline beliefs about employer response. Panel A reports the heterogeneous treatment effects for feedback-only and control groups and Panel B on employer info-only and control groups. All regressions control for round fixed effects. The omitted group is low-type workers in the control group. Standard errors are clustered at the job level. $p < 0.10^*$, $p < 0.05^{**}$, $p < 0.01^{***}$.

Appendix B Reputation Model with Belief Frictions

B1 Setup

Consider a labor market with a unit mass of workers and employers. Workers, indexed by i , live for two periods, while employers, indexed by j , live for one period. Workers hired in the first period become *veterans* in the second period, while those in their first period or not hired previously are referred to as *novices*. At the end of each period, all employers and second-period workers exit the market and are replaced by a new generation.

At the beginning of the period, each employer posts an identical job on the online platform with an exogenous budget $\bar{w}_j$. The budget $\bar{w}_j$ represents the maximum amount the employer is willing to pay for an online freelancer, based on the cost of hiring a local worker offline.

Workers differ in their innate ability $\theta_i \in \{H, L\}$ to execute the task. We assume that workers know their own type upon entering the market. Workers decide whether to apply to a job by submitting a wage bid $w_{ij} \leq \bar{w}_j$ or not applying at all.²⁴ If hired, the worker is paid w_{ij} ; otherwise, the worker receives the outside option $c_\theta = \{c_H, c_L\}$.

Employers derive value $v_\theta = \{v_H, v_L\}$ from a worker's output but cannot observe the worker type prior to hiring. Instead, they observe proposed wages for all applicants and public reviews r_i for veteran workers. For veterans, employers infer their type from their review history. For novices, employers form beliefs about their type based on their proposed wages w_{ij} : a novice is high type with probability $\gamma(w_{ij})$ and low type with probability $1 - \gamma(w_{ij})$. If all novices propose the same wage, or if wages are uninformative about worker type, employers may screen candidates through interview at costs s_j and obtain additional quality cues λ_{ij} . The employer's objective is to maximize the expected return from hiring:

$$\max_i \{E[v_i | r_i, w_{ij}, \lambda_{ij} \times 1[\text{screening}]] - w_{ij} - s_j \times 1[\text{screening}], \bar{v}_j\}$$

where $\bar{v}_j$ is the employer's outside option. Employers hire from the platform only if expected returns exceed $\bar{v}_j$.

Once a task is completed, employers observe the output and leave a public rating on the platform that reveals the worker's type. Following a bad review, low-type veterans will not receive repeated offers in the next period. High-type veterans, in contrast, will be hired again at $\bar{w}_j \in \{c_H, v_H\}$. The worker's problem is therefore to set wages in the novice period to maximize expected lifetime earnings on the platform:

$$\max_{w_{ij}} (w_{ij} - c_i) + \beta(\bar{w}_j - c_i) \times 1[\theta_i = H]$$

The timeline of events within each period is as follows:

- (i) A new generation of workers and employers joins the platform. Each employer posts a job with budget $\bar{w}_j$.
- (ii) Workers observe $\bar{w}_j$ and decide whether to apply by proposing a wage w_{ij} .

²⁴This is empirically supported: fewer than 3% freelancers in our application data proposed wages higher than the budget.

- (iii) Employers observe the proposed wage w_{ij} and public reviews r_i if they exist, and possibly incur s_j to screen inexperienced candidates. They make at most one offer to the candidate that generates the highest return.
- (iv) Hired workers produce the output and their type is revealed.
- (v) All employers and workers in their second period exit the market.

B2 Market Equilibrium

We examine labor market outcomes under the three scenarios: (i) the benchmark case as in [Tirole \(1988\)](#), (ii) novices having incomplete information about own ability, (iii) novices believing that employers interpret low wage offers as signals for low quality for novices. The experimental treatments are designed to shift employers and workers across these scenarios.

Proposition 1. *In the benchmark case, a perfect Bayesian equilibrium exists when returns to reputation for high-types exceed the difference in outside option between types: $\beta(\bar{w}_j - c_H) \geq c_H - c_L$. High-type novices propose wages below the job budget at $w_{Hj} = c_L < v_H$ in period 1, while low-types propose $\bar{w}$. Employers interpret low wages as credible quality signals and hire all high-type novices and veterans in equilibrium.*

Proof. Even though employers cannot observe the type of novice workers prior to hiring, high-type novices may distinguish themselves from low-type peers by proposing lower wages in period 1 as an investment in their reputation to generate repeated hiring on the platform. That is, high-type novices offer wages w_{Hj} to maximize their lifetime earnings:

$$\max_{w_{Hj}} \Pi_H(w_{Hj}) = (w_{Hj} - c_H) + \beta(\bar{w}_j - c_H)$$

The optimization is subject to the following constraints:

$$\begin{cases} \Pi_L(w_{Hj}^*) = (w_{Hj}^* - c_L) \leq 0 & \text{Incentive Compatibility (IC)} \\ \Pi_H(w_{Hj}^*) = (w_{Hj}^* - c_H) + \beta(\bar{w}_j - c_H) \geq 0 & \text{Participation Constraint (IR)} \end{cases}$$

High-type novices set wages equal to the low-type's outside option to ensure that at w_{Hj}^* , low-type novices will incur a loss in lifetime earnings, but not themselves. Together, these two constraints imply that wage separation by types can be sustained if for high-type novice workers, returns to reputation for high-type novices exceeds the cost differential between types: $\beta(\bar{w}_j - c_H) \geq c_H - c_L$.

Consistent with this wage-setting strategy, employers can infer high quality from low wages proposed by novices such that $\gamma'(w_{ij}) < 0$. No costly screening is required. Hence, in equilibrium, employers hire all high-type novices at $w_{Hj}^* = c_L$ and veterans.

Proposition 2. *When workers have limited information on own type, there is no complete separation in proposed wages by unobserved type. Employers rely on costly screening to distinguish among novice workers. In equilibrium, novices proposing $w_{ij} < \lambda_{ij}v_H + (1 - \lambda_{ij})v_L - s_j$ and high-type veterans are hired. Compared to the benchmark case, fewer novices are employed and average output quality is lower.*

Proof. Suppose neither workers nor employers know worker type upon entering the market. Workers believe that they belong to the high type with probability μ_i and the low type with probability $1 - \mu_i$. Once hired, they learn their true type by observing public reviews.

Under imperfect information about their own types, novice workers propose wages w_{ij} to maximize their expected lifetime earnings given their beliefs μ_i :

$$\max_{w_{ij}} \Pi(w_{ij}, \mu_i) = w_{ij} - (\mu_i c_H + (1 - \mu_i) c_L) + \beta \cdot \mu_i (\bar{w}_j - c_H)$$

In this case, proposing lower wages no longer signals true high ability but rather reflects novice workers' belief about being the high-type. Consequently, employers no longer view proposed wages as credible signals of novices' ability and therefore incur costly screening to discern types through the additional signal λ_{ij} . Novice workers may still be hired if they can compensate for the employer's screening cost by offering lower wages. That is

$$\lambda_{ij} v_H + (1 - \lambda_{ij}) v_L - w_{ij} - s_j \geq 0$$

However, due to the additional screening costs, employers hire fewer novices in equilibrium and average output quality is lower than the benchmark case.

Proposition 3. *When employers interpret low wage offers by novices as signals of low quality, labor demand is more responsive to undercutting by veterans than by novices. If quality concerns dominate cost considerations in hiring decisions, novice workers pool at the job budget $\bar{w}_j < \lambda_{ij} v_H + (1 - \lambda_{ij}) v_L - s_j$ and high-type veterans with $\bar{w}_j \in \{c_H, v_H\}$ are hired. Compared to the benchmark case, hiring novices becomes more costly and fewer are employed.*

Proof. Now suppose employers interpret higher proposed wages as signals of higher quality, even in the absence of other credible signals: $\gamma(w_{ij})$ is increasing in w_{ij} .

Under this condition, demand for novices is less responsive to proposed wages than demand for veterans. If workers offer lower wages, this reduces hiring costs for both novices and veterans, but also reduces perceived returns for novices only. Whether proposing lower wages improves novices' hiring chances depends on the relative strength of the cost and perceived quality channels.

When the cost channel dominates, high-type novice workers can still increase their hiring probability by proposing lower wages as in the standard case. Once employers hire novice workers who undercut and discover their type, they can update their beliefs about $\gamma(w_{ij})$ and thus, employers' perception will be corrected over time.

When quality concerns dominate, novice workers avoid undercutting due to concerns about negative signals and propose wages equal to the job budget, $w_{ij} = \bar{w}_j$. Without variation in proposed wages, employers need to incur screening cost s_j to obtain additional quality cues λ_{ij} . Since novice workers with unknown type charge the same as veteran workers in this case, employers will always prefer the experienced ones. Nevertheless, novices may be hired if the expected returns can compensate for employers' screening costs. That is

$$\lambda_{ij} v_H + (1 - \lambda_{ij}) v_L - \bar{w}_j - s_j \geq 0$$

As employers do not observe proposed wages below $\bar{w}_j$, they cannot update their beliefs about $\gamma(w_{ij})$ over time. In equilibrium, employers hire fewer novices and average output quality is lower than the benchmark case.